EtherCAT® Implementation Directive for CiA402 Drive Profile

Directive for using IEC 61800-7-201 within EtherCAT-based servo drives

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DOCUMENT HISTORY

Version	Comment
1.0.0 Release	Homing Method 35 obsolete, Method 37 added instead
1.1.0 Release	5.4 Introduce Process Data state 6.2, 6.3, 6.4, 6.5: Add Input / Output cycle counter in control/status word 6.6: Add reference to profile position mode 6.7 Corrections of Polarity object ($607E_h$), adaptions also in csp, csv, cst mode figures 11.1: Example of PDO Mapping for dynamic switching of operation modes 13: Add stepper mode specific clause 14: Guidance for drive use cases 15: Extend “Device Type” with information about supporting of ETG.6100 Safety Drive Profile 16: Delete extension to up to 16 axes, since manufacturer specific area cannot be extended

CONTENTS

LEGAL NOTICE DOCUMENT HISTORY 2
1 Scope 9
1.1 IEC 61800-7 10
2 References 11
3 Terms and Definitions 12
4 Error codes and error behavior 13
5 Controlling the power drive system 14
5.1 State machine 14
5.2 Controlword (Object 6040_h) 15
5.3 Statusword (Object 6041_h) 16
5.4 Process Data state 16
5.4.1 Object $603E_h$: TxPDO Data Invalid 17
6 Modes of Operation 18
6.1 General 18
6.2 Cyclic synchronous position mode (csp) 19
6.3 Cyclic synchronous velocity mode (csv) 21
6.4 Cyclic synchronous torque mode (cst) 23
6.5 Cyclic synchronous torque mode with commutation angle (cstca) 25
6.5.1 General information 25
6.5.2 General definitions 26
6.5.3 Functional description 26
6.5.4 Object $60EA_h$: Commutation angle 27
6.6 Profile Position mode (pp) 28
6.7 Polarity object 28
6.8 Object definitions 28
6.8.1 Object 6502_h: Supported drive modes 28
6.8.2 Object $60D9_h$: Supported synchronization functions 29
6.8.3 Object $60DA_h$: Synchronization function settings 30
6.9 Switching between Operation modes 32
6.9.1 Static setting of operation mode 32
6.9.2 Dynamic switching of operation mode in state operation 32
7 Function Groups 34
7.1 General 34
7.2 Function Group "Torque Limiting" 34
7.2.1 Object $60E0_h$: Positive torque limit value 34
7.2.2 Object $60E1_h$: Negative torque limit value 35
7.3 Homing 36
7.3.1 Homing Method 35: Homing on current position (obsolete) 36

7.3.2 Homing Method 36: Homing with touch-probe – reserved for compatibility reasons 36
7.3.3 Homing Method 37: Homing on current position 36
7.3.4 Calculation of position actual value by homing process 36
7.3.5 Homing object list 37
7.3.6 Object 60E3$_h$: Supported Homing Methods 37
7.4 Function Group "Touch Probe" 39
7.4.1 Object 60B8 $_h$: Touch probe function 40
7.4.2 Object 60D0$_h$: Touch probe source 40
7.4.3 Object 60B9$_h$: Touch probe status 42
7.4.4 Touch probe time stamp latch 44
7.4.5 Touch probe edge counter for continuous mode 46
7.4.6 Application Example: Controller based “homing” with touch probe 48
8 Factor Group 50
9 Support of additional sensor interfaces 51
9.1.1 Object 60E4$_h$: Additional position actual value 52
9.1.2 Object 60E5$_h$: Additional velocity actual value 53
9.1.3 Object 60E6$_h$: Additional position encoder resolution – encoder increments 54
9.1.4 Object 60EB$_h$: Additional position encoder resolution – Motor revolutions 55
9.1.5 Object 60E7$_h$: Additional velocity encoder resolution - Encoder increments per second 56
9.1.6 Object 60EC$_h$: Additional velocity encoder resolution – Motor revolutions per second 57
9.1.7 Object 60E8$_h$: Additional gear ratio – Motor revolutions 58
9.1.8 Object 60ED$_h$: Additional gear ratio – Shaft revolutions 59
9.1.9 Object 60E9$_h$: Additional feed constant – Feed 60
9.1.10 Object 60EE$_h$: Additional feed constant – Shaft revolutions 61
10 Endless positioning 62
11 Process data objects (PDO) 64
11.1 Example of PDO mapping implementations for dynamic switching of operation modes 65
12 Synchronization 66
13 Stepper Motor specifics 67
13.1 Basic assumption for stepper drives in this directive 67
13.2 Resolution factor 67
13.2.1 Object 60EF$_h$: Motor resolution 68
14 Guidance for drive use cases 69
15 Device Type 70
16 Object list 71

TABLES

Table 1: Supported Error option codes 13
Table 2: State Machine - States 15
Table 3: Use of controlword 15
Table 4: Use of statusword 16
Table 5: Object description object $603E_h$ 17
Table 6: Entry description object $603E_h$ 17
Table 7: Operation Modes 18
Table 8: Modes of operation – Object list 18
Table 9: csp mode – Object list 19
Table 10: csv mode – Object list 21
Table 11: cst mode – Object list 23
Table 12: cstca mode – Object list 26
Table 13: Object description 27
Table 14: Entry description 27
Table 15: pp mode – Object list 28
Table 16: Object description 28
Table 17: Entry description 28
Table 18: Object description 29
Table 19: Entry description 29
Table 20: Object description 30
Table 21: Entry description 30
Table 22: FG Torque Limiting – Object list 34
Table 23: Object description object $60E0_h$ 34
Table 24: Entry description object $60E0_h$ 34
Table 25: Object description object $60E1_h$ 35
Table 26: Entry description object $60E1_h$ 35
Table 27: Homing methods 36
Table 28: Homing – Object list 37
Table 29: Object description object $60E3_h$ 37
Table 30: Entry description object $60E3_h$ 38
Table 31: FG Touch probe – Object list 39
Table 32: Touch Probe objects for optional Input 2 39
Table 33: Value definition object $60B8_h$ 40
Table 34: Value definition 40
Table 35: Object description 41
Table 36: Entry description 41
Table 37: Value definition object $60B9_h$ 42
Table 38: Touch Probe Timing example 43
Table 39: Object description 44
Table 40: Entry description 44
Table 41: Object description 44
Table 42: Entry description 44
Table 43: Object description 45
Table 44: Entry description 45
Table 45: Object description 45
Table 46: Entry description 45
Table 47: Object description 46
Table 48: Entry description 46
Table 49: Object description 46
Table 50: Entry description 47
Table 51: Object description 47
Table 52: Entry description 47
Table 53: Object description 47
Table 54: Entry description 47
Table 55: Steps for control device based homing 48
Table 56: Default values for Units 50
Table 57: Additional sensor information – Object list 51
Table 58: Object description object $60E4_h$ 52
Table 59: Entry description object $60E4_h$ 52
Table 60: Object description object $60E5_h$ 53

Table 61: Entry description object x60E5 53
Table 62: Object description object $60E6_h$ 54
Table 63: Entry description 54
Table 64: Object description object $60EB_h$ 55
Table 65: Entry description 55
Table 66: Object description object $60E7_h$ 56
Table 67: Entry description 56
Table 68: Object description object $60EC_h$ 57
Table 69: Entry description 57
Table 70: Object description object $60E8_h$ 58
Table 71: Entry description 58
Table 72: Object description object $60ED_h$ 59
Table 73: Entry description 59
Table 74: Object description object $60E9_h$ 60
Table 75: Entry description 60
Table 76: Object description object $60EE_h$ 61
Table 77: Entry description 61
Table 78: Value definition for bit 6 and 7 of object $60F2_h$ 63
Table 79: Default PDOs 64
Table 80: Recommended PDO Mapping according to supported modes of operation 65
Table 81: Object description object $60EF_h$ 68
Table 82: Entry description 68
Table 83: Drive use cases regarding to PLC functionality 69
Table 84: Object 1000_h Device Type for EtherCAT servo drive 70
Table 85: Object dictionary – object list (numerical order) 71

FIGURES

Figure 1: Structure of the IEC 61800 - 7 10
Figure 2: State Machine 14
Figure 3: Cyclic synchronous position mode overview 19
Figure 4: Cyclic synchronous position control function 19
Figure 5: Statusword for *csp* mode 20
Figure 6: Cyclic synchronous velocity mode overview 21
Figure 7: Cyclic synchronous velocity control function 21
Figure 8: Statusword for *csv* mode 22
Figure 9: Cyclic synchronous torque mode overview 23
Figure 10: Cyclic synchronous torque control function 23
Figure 11: Statusword for *cst* mode 24
Figure 12: Cyclic synchronous torque mode with commutation angle overview 25
Figure 13: Cyclic synchronous torque with commutation angle control function 26
Figure 14: Statusword for *cstca* mode 26
Figure 15: Value definition 28
Figure 16: Value definition 29
Figure 17: Value definition 30
Figure 18: Dynamic change of operation mode 33
Figure 19: Usage of torque limiting objects 34
Figure 20: Home Offset definition 37
Figure 21: Object $60E3_h$ Supported Homing Methods – Value definition 38
Figure 22: Timing diagram for Touch probe example 43
Figure 23: Control device based homing with touch probe 48
Figure 24: Position actual value in modulo format 62
Figure 25: Object structure $60F2_h$ 63
Figure 26: Rotary axis positioning 63
Figure 27: Stepper drive without feedback system 67

1 Scope

IEC 61800–7–201 and IEC 61800–7–301 specify the CiA402 drive profile which is mapped to EtherCAT. In the IEC Standard many objects and operation modes are defined as optional therefore this document intends to define a common behavior of an EtherCAT servo drive supporting the CiA402 drive profile.

Scope of this directive

- CiA402 Servo drives with EtherCAT interface

Frequency converters are intentionally not within the scope of this directive.

Special behavior for stepper motors is defined in clause 13.

The mapping of the SERCOS profile to EtherCAT described in IEC 61800-7-304 is not part of this directive. The specific usage of this part of the standard is defined within the SERCOS International e.V.

NOTE Objects / functions / operation modes in bold letters throughout the text and the figures means that this feature is mandatory according to this directive.

This directive is not a substitute of the IEC 61800-7-201 specification! For the understanding and implementation of the drive profile the IEC specification is indispensable.

The directive gives explanations and defines enhancements according to the IEC specification.

1.1 IEC 61800-7

IEC 61800-7 specifies profiles for power drive systems (PDS) and their mapping to existing communication systems by use of a generic interface model.

IEC 61800-7-1 specifies a generic interface between power drive system(s) (PDS) and the application control program in a controller. The generic PDS interface is not specific to any particular communication network technology. Annexes of IEC 61800-7-1 specify the mapping of the different drive profiles onto the generic PDS interface.

IEC 61800-7-2xx specifies different profiles for power drive systems (PDS). The profiles of the PDS are not specific to any particular communication network technology. This directive deals with subpart -201 that specifies the CiA402 DriveProfile.

IEC 61800-7-3xx specifies the mapping of different profiles to different network technologies.

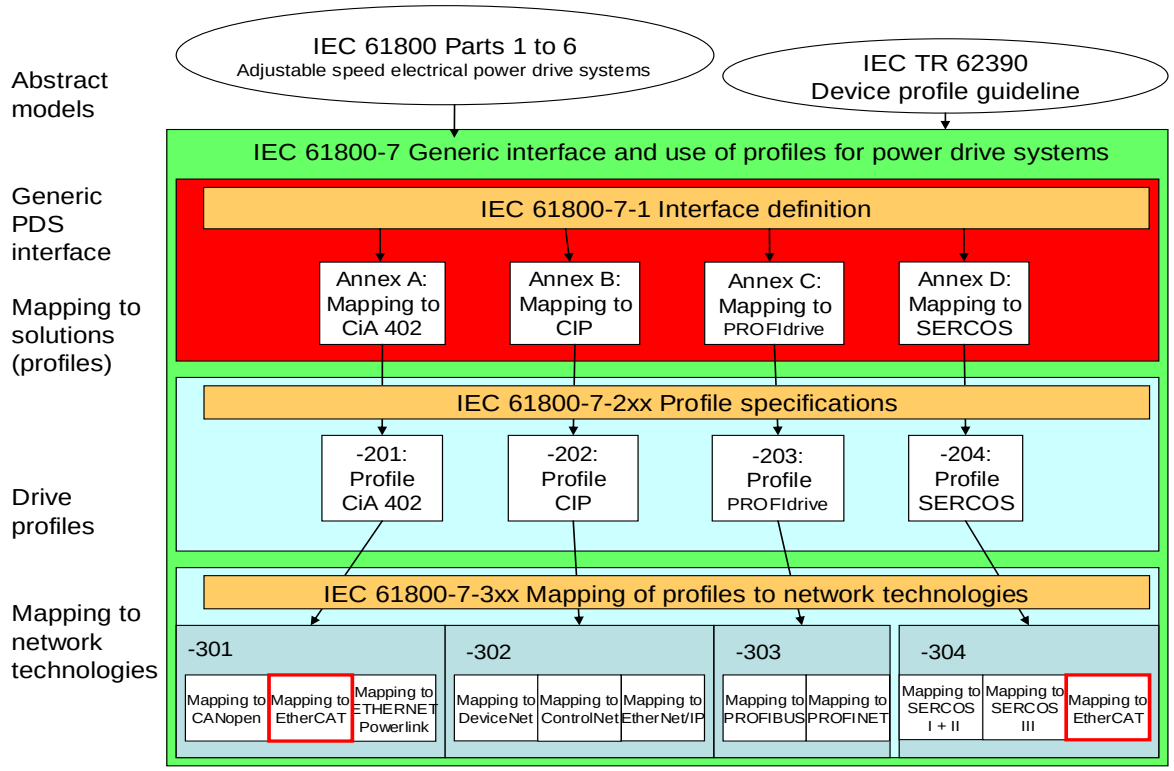


Figure 1: Structure of the IEC 61800 - 7

2 References

[1] IEC 61800 (all parts), Adjustable speed electrical power drive systems

[2] IEC 61800-7-1:—, Adjustable speed electrical power drive systems – Part 7-1 Generic interface and use of profiles for power drive systems – Interface definition

[3] IEC 61800-7-201:—, Adjustable speed electrical power drive systems – Part 7-2 Generic interface and use of profiles for power drive systems – Profile specification Type 1

[4] IEC 61800-7-3:—, Adjustable speed electrical power drive systems – Part 7-2 Generic interface and use of profiles for power drive systems – Mapping of profiles to network technologies

[5] IEC 61158-3-12, Digital data communications for measurement and control – Fieldbus for use in industrial control systems – Part 3-12: data-link layer service definition – Type 12 elements

[6] IEC 61158-4-12, Digital data communications for measurement and control – Fieldbus for use in industrial control systems – Part 4-12: Data-link layer protocol specification – Type 12 elements

[7] IEC 61158-5-12, Digital data communications for measurement and control – Fieldbus for use in industrial control systems – Part 5-12: Application layer service definition – Type 12 elements

[8] IEC 61158-6-12, Digital data communications for measurement and control – Fieldbus for use in industrial control systems – Part 6-12: Application layer protocol specification – Type 12 elements

[9] ETG.1020, EtherCAT Guidelines & Protocol Enhancements.doc, ETG guideline for synchronization

[10] ETG.6100 Safety Drive Profile series

3 Terms and Definitions

(M) Mandatory
the object / feature shall be supported according to IEC 61800-7 or according to this directive.

(C) Conditional
the object / feature parameter is conditional upon other parameters / features. I.e. it shall be supported if the other parameter / feature is supported.

(R) Recommended
the object / feature is highly recommended according to this directive.
It is defined for a better quality of the corresponding function.

(O) Optional
the object / feature might be supported.
It is not recommended according to this directive.

(FG) Function Group
combination of objects and features to support a specific function within the drive.

Low-level power
electrical power supply for the control section of the drive, e.g. 24 V.

High-level power
main electric power supply of the drive, e.g. 230 V or 380 V.

4 Error codes and error behavior

The following error option codes are supported:

Table 1: Supported Error option codes

Index	Name	Category	Remark
6007_h	Abort connection option code	O	
$605A_h$	Quick stop option code	O	
$605B_h$	Shutdown option code	O	if there is a transition from Operation Enabled state to Ready To Switch On state. The slow down ramp is the deceleration value of the used mode of operations.
$605C_h$	Disable operation option code	O	if there is a transition from Operation Enabled state to Switched on state. The slow down ramp is the deceleration value of the used mode of operations
$605D_h$	Halt option code	O	
$605E_h$	Fault reaction option code	O	when a fault is detected in the PDS. The slow down ramp is the deceleration value of the used mode of operations. Default: +2 (quick stop ramp)

The communication state machine (ESM) is independent of the drive State machine (FSA SM).

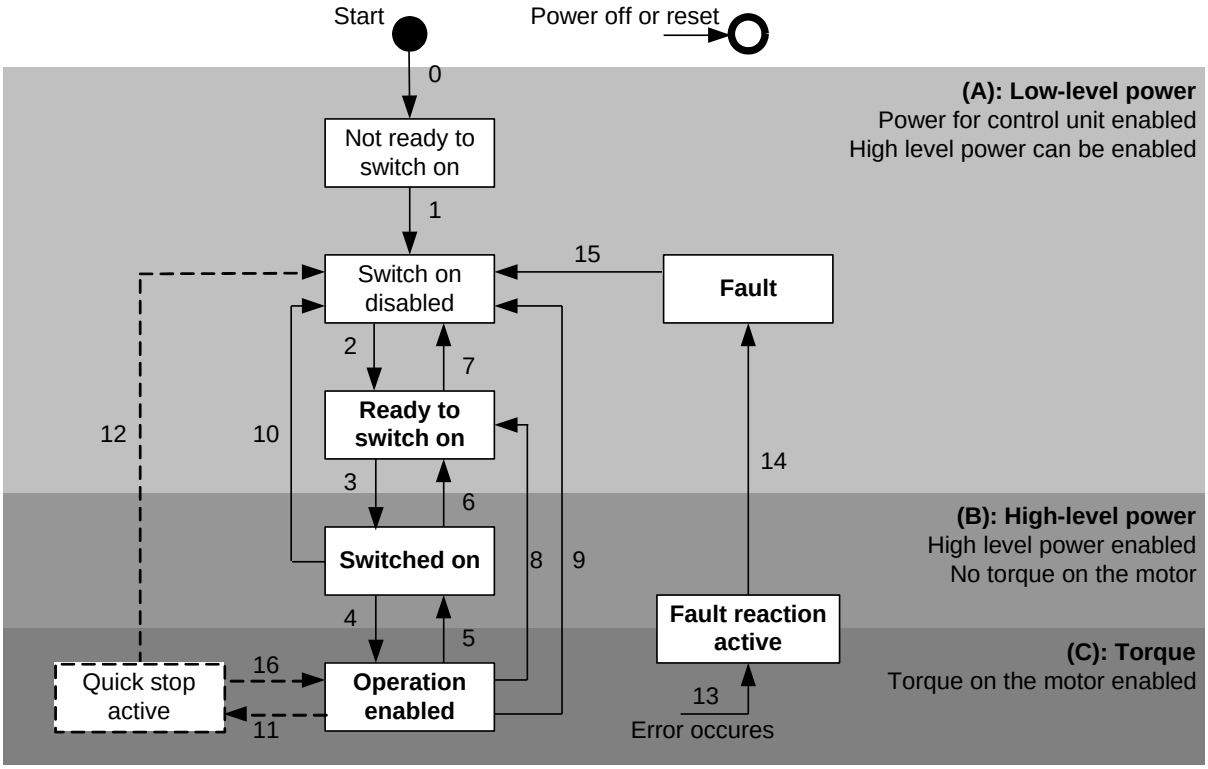
Object 6007_h can be used to define the behavior in case the slave ESM leaves the OP state.

A state change of the FSA SM does not affect the ESM.

NOTE: See also [9]: Error Handling, Object $10F1_h$ Error Settings for multiple Synch Error behavior.

5 Controlling the power drive system

5.1 State machine



state — Optional state

state — State can be changed manually by the slave

state — State is checked by master

Figure 2: State Machine

The state *Quick stop active* is optional. Usually the control device will perform the quick stop function.

The state *Not ready to switch on* is performed automatically by the drive.

The state *Switch on disabled* can be passed through automatically by local signals of the drive. The control device has to compare during start-up which state is reached by the drive. Then the control device has to perform the transition *2* only if necessary.

The states in bold letters are stable states in the drive and will be checked by the control device. The transitions *3* and *4* can only be requested by the control device.

If the control device sets Bits 0, 1 and 3 simultaneous in the controlword the drive can pass from *Ready to Switch on* via *Switched on* to *Operation enabled* with one control device command.

In area (A) the Low-level power is enabled. The High-level power may be enabled, e.g. to supply the low-level power.

In area (B) the High-level power shall be switched on but there is no torque on the motor. Transition *3* shall be refused by the drive, if no high-level power is enabled. Target and set-point values shall be ignored.

In area (C) the drive is ready to operate and the torque is switched on to the motor. Target and set-point values shall be processed.

During transition *4* the torque can be set up and if necessary a brake can be opened. The statusword displays the state *Switched on* until the drive is ready to follow the target and set-point values.

Accordingly in transition 5 the drive can actively be decelerated and the brake can be closed. The statusword displays the state Operation enabled as long as torque on the motor is enabled.

The servo drive shall support the functions as shown in Table 2.

Table 2: State Machine - States

Function	FSA states							
	Not ready to switch on	Switch on disabled	**Ready to switch on**	**Switched on**	**Operation enabled**	Quick stop active	**Fault reaction active**	**Fault**
	O	O	**M**	**M**	**M**	O	**M**	**M**
Brake applied, if present	Yes	Yes	Yes	Yes	Yes/No	Yes/No	Yes/No	Yes
Low-level power applied	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
High-level power applied	Yes/No	Yes/No	Yes/No	Yes	Yes	Yes	Yes	Yes/No
Drive function enabled	No	No	No	No	Yes	Yes	Yes	No
Configuration allowed	Yes	Yes	Yes	Yes	Yes/No	Yes/No	Yes/No	Yes

5.2 Controlword (Object 6040_h)

The servo drive shall support the following Bits in the *controlword*.

Table 3: Use of controlword

Bit	Meaning	Category
0	**Switch on**	**M**
1	**Enable voltage**	**M**
2	Quick stop	O
3	**Enable operation**	**M**
4 – 6	Operation mode specific	O
7	**Fault reset**	**M**
8	Halt	O
9	Operation mode specific	O
10	reserved	O
11 – 15	Manufacturer specific	O

Bit 2 *Quick stop* shall be set (= 1) by the control device to deactivate an optional quick stop. This bit shall be ignored by a drive that does not support the quick stop state.

5.3 Statusword (Object 6041_h)

The servo drive shall support the following Bits in the *statusword*.

Table 4: Use of statusword

Bit	Meaning	Category	
0	**Ready to switch on**	**M**	
1	**Switched on**	**M**	
2	**Operation enabled**	**M**	
3	**Fault**	**M**	
4	Voltage enabled	O	
5	Quick stop	O	
6	**Switch on disabled**	**M**	
7	Warning	O	
8	Manufacturer specific	O	
9	Remote	O	
10	Operation mode specific	O	
11	Internal limit active	O	
12	Operation mode specific	C	Mandatory for csp, csv, cst mode
13	Operation mode specific	O	
14 – 15	Manufacturer specific	O	

Bit 5 *Quick Stop* shall be set by the drive if the Quick Stop state is not supported or the Quick stop function is not active.

Bit 11 *Internal limit active* is set if internal limits are exceeded so that the target and set-point values can't be reached. E.g. for hardware position switches, current limiter or thermal overload.

5.4 Process Data state

The control unit needs information about the validity of the current process data. Even if the drive is in Fault state the process data, e.g. position actual value, may still be valid and can be used be the control unit application.

EXAMPLE: in a coupled master / slave system, the master axis might need to go to FAULT state, but its current position information is still valid. The motion controller can use the valid master's position for the slave axis so that they still can follow.

5.4.1 Object $603E_h$: TxPDO Data Invalid

This object defines the current validity state of a TxPDO. Each entry represents the state of a corresponding PDO. The entry shall be set (TRUE) if the process data of the corresponding TxPDO is not valid.

Table 5 specifies the object description, and Table 6 specifies the entry description.

Table 5: Object description object $603E_h$

Attribute	Value
Index	$603E_h$
Name	TxPDO Data invalid
Object Code	RECORD
Category	Optional

Table 6: Entry description object $603E_h$

Attribute	Value
Sub-Index	00_h
Description	Highest sub-index supported
Entry category	Mandatory
Access	constant
PDO Mapping	no
Value Range	1 to 254
Default Value	Manufacturer-specific
Sub-Index	01_h
Description	TxPDO Data Invalid of 1st PDO
Entry category	Mandatory
Access	Ro
PDO Mapping	Yes
Data Type	BOOL
Default Value	No
Sub-Index	02_h
Description	TxPDO Data Invalid of 2nd PDO
Entry category	Optional
Access	ro
PDO Mapping	yes
Data Type	BOOL
Default Value	No
	to
Sub-Index	FE_h
Description	TxPDO Data Invalid of 254th PDO
Entry category	Optional
Access	ro
PDO Mapping	yes
Data Type	BOOL
Default Value	No

6 Modes of Operation

6.1 General

The following operation modes for the servo drives are recommended:

Table 7: Operation Modes

Mode of operation	Abbr.	Code	Category	Remarks
Profile position mode	pp	1	O	
Velocity mode (frequency converter)	vl	2	O	
Profile velocity mode	pv	3	O	
Torque profile mode	tq	4	O	
Homing mode	hm	6	O	
Interpolated position mode	ip	7	O	
Cyclic synchronous position mode	**csp**	**8**	**C**	at least one of these modes shall be supported
Cyclic synchronous velocity mode	**csv**	**9**	**C**	
Cyclic synchronous torque mode	**cst**	**10**	**C**	
Cyclic synchronous torque mode with commutation angle	cstca	11	O	
Manufacturer specific mode		-128…-1	O	

At least one of the cyclic synchronous modes shall be supported. The csp mode is the most common mode supported by the control units.

Table 8: Modes of operation – Object list

Index	Name	Category
6060_h	**Modes of operation**	**M**
6061_h	**Modes of operation display**	**M**
6502_h	Supported drive modes	M

6.2 Cyclic synchronous position mode (csp)

Figure 3 shows an overview of the *Cyclic synchronous position mode (csp)*.

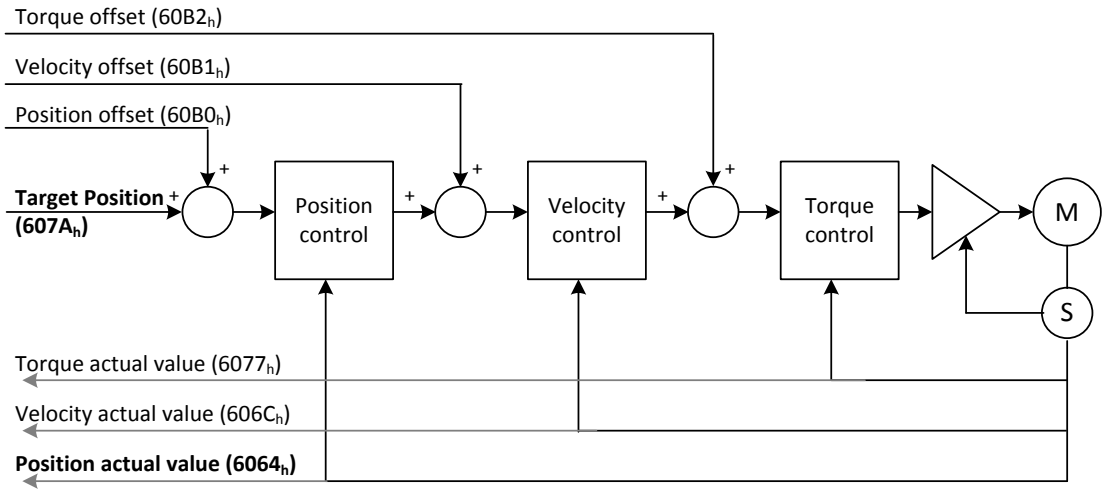


Figure 3: Cyclic synchronous position mode overview

The following (bold) objects are used for *csp* (see Figure 4).

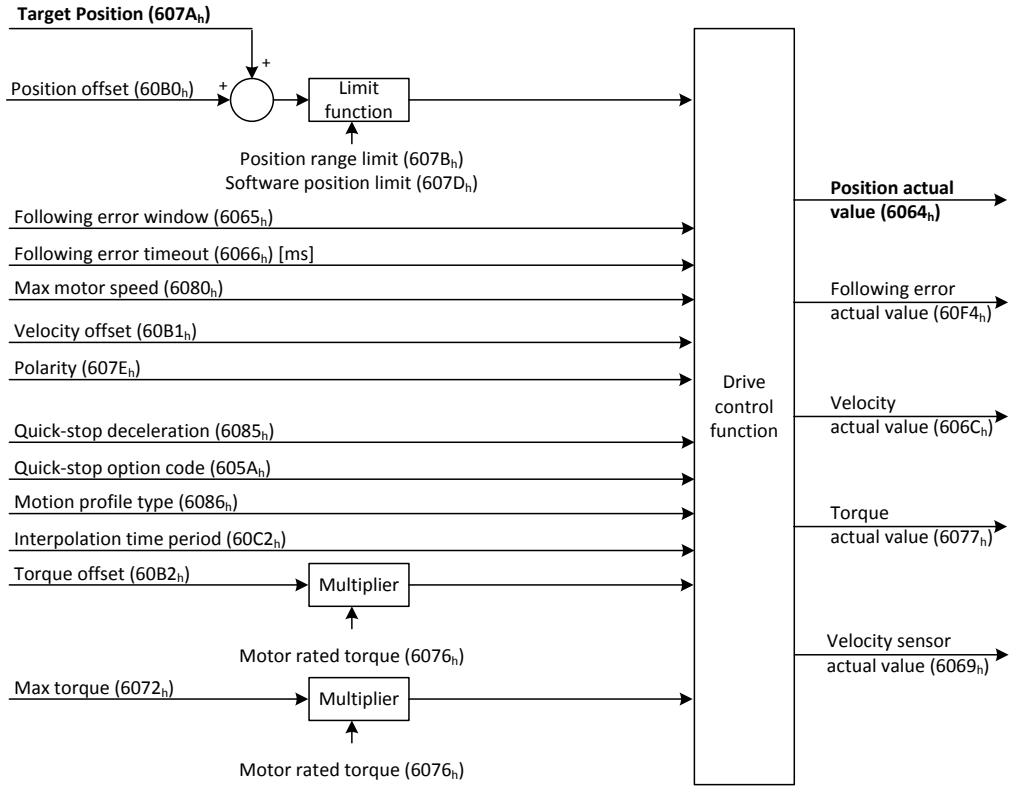


Figure 4: Cyclic synchronous position control function

Table 9: csp mode – Object list

Index	Name	Category
$607A_h$	**Target Position**	**M**
6064_h	**Position actual value**	**M**
6077_h	Torque actual value	R
$607B_h$	Position range limit	R
$607D_h$	Software position limit	R

Index	Name	Category
$60B1_h$	Velocity offset	R
$60B2_h$	Torque offset	R
$60F4_h$	Following error actual value	R
6065_h	Following error window	R
6066_h	Following error time out	R
$60C2_h$	Interpolation time period	R

If the following error is calculated in the control device it is afflicted with a dead-time. Therefore the calculation of the following error in the drive might have a better quality.

The following error value shall only be evaluated in state *Operation enabled*.

After a Reset the set point should be set to the actual value so that the following error is zero.

In the *csp* mode the Halt bit (bit 8) of the controlword shall be ignored because the halt function is controlled by the control device.

Bit 5 and 6 of the controlword can be used as Output Cycle Counter. This 2-Bit counter can be used by the control device to indicate if updated output data are available. The counter shall be incremented with every update of the output process data. Object $60D9_h$ and $60DA_h$ shall be supported and used to enable or disable the Output Cycle Counter functionality.

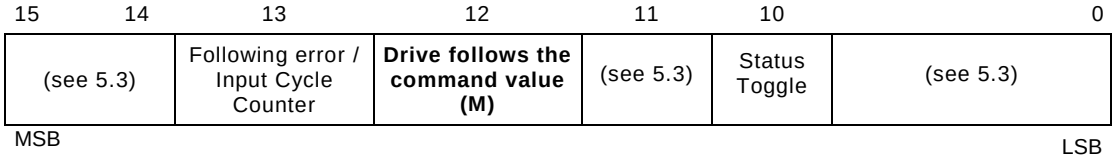


Figure 5: Statusword for *csp* mode

In the Statusword Bit 12 is mandatory.

The bit 10 is used in Profile position mode as "target reached" information. In *csp* the new target position is given cyclically by the control device. This bit can be used as *Status Toggle* information to indicate if the device provides updated input data. The bit shall be toggled with every update of the input process data. If object $60D9_h$ is supported, the Status Toggle function can be enabled or disabled.

The Following Error functionality is usually not supported by the drive in csp mode therefore Bit 13 can be used to extend the Status Toggle information to a 2-Bit Input Cycle Counter. Object $60D9_h$ and $60DA_h$ shall be supported and used to enable or disable the Input Cycle Counter functionality.

The Bit 12 *Drive follows the command value* shall be zero if the drive **does not** follow the target value (position, velocity or torque) because of local reasons (internal set-point settings), e.g. if a local Input is configured to a halt function or a safety function prevents the drive in Operational to follow the target set point. The control device shall evaluate the bit. The Bit 12 shall be set if the drive is in state *Operation enabled* and follows the target and set-point values of the control device. In all other cases it shall be zero. If the bit is not supported it shall be fix set to 1 in the statusword.

NOTE: If Bit 12 is not supported and is not set to 1, the control device will not run the drive, because it expects that the drive does not follow the command.

6.3 Cyclic synchronous velocity mode (csv)

Figure 6 shows an overview of the *Cyclic synchronous velocity mode (csv)*.

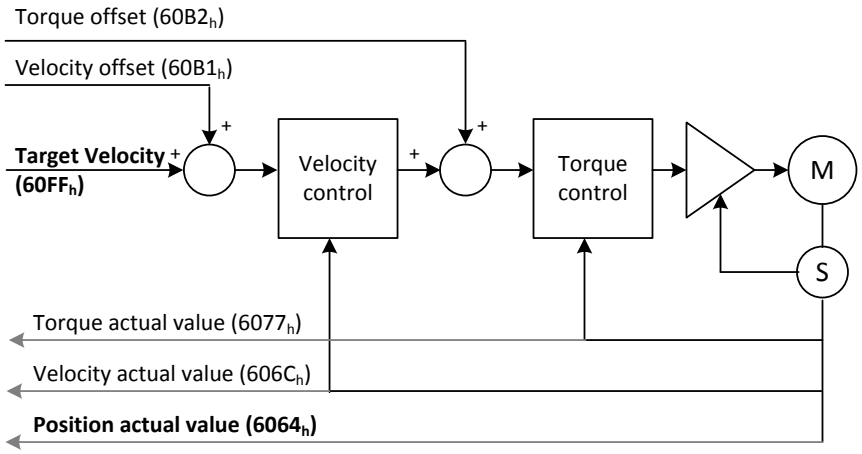


Figure 6: Cyclic synchronous velocity mode overview

The following (bold) objects are used for *csv* (see Figure 7).

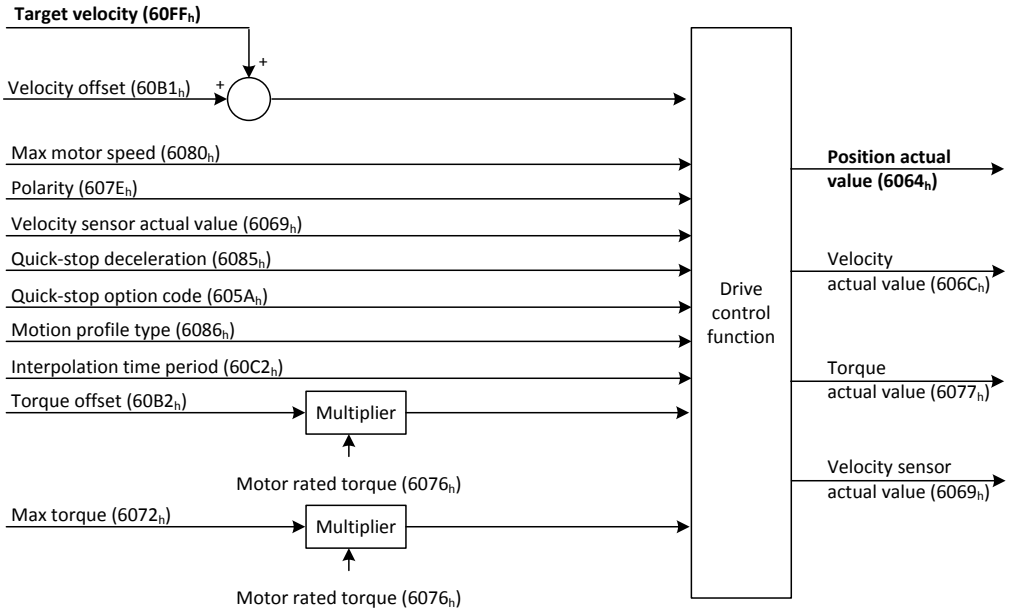


Figure 7: Cyclic synchronous velocity control function

Table 10: csv mode – Object list

Index	Name	Category
60FF$_h$	**Target Velocity**	**M**
6064$_h$	**Position actual value**	**M**
606C$_h$	Velocity actual value	R
6077$_h$	Torque actual value	R
60B2$_h$	Torque offset	R
60C2$_h$	Interpolation time period	R

NOTE: in CiA-402 specification the velocity actual value is used as mandatory output to the control device.

In the csv mode the Halt bit (bit 8) of the Controlword shall be ignored because the halt function is controlled by the control device.

Bit 5 and 6 of Controlword can be used as Output Cycle Counter. This 2-Bit counter can be used by the control device to indicate if updated output data are available. The counter shall be incremented

with every update of the output process data. Object $60D9_h$ and $60DA_h$ shall be supported and used to enable or disable the Output Cycle Counter functionality.

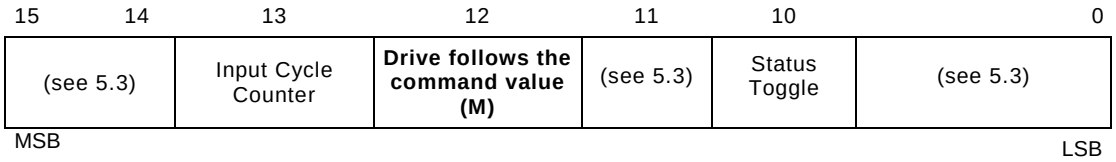


Figure 8: Statusword for *csv* mode

In the Statusword Bit 12 is mandatory.

In *csv* mode Bit 10 can be used as *Status Toggle* information to indicate if the device provides updated input data. The bit shall be toggled with every update of the input process data. If object $60D9_h$ is supported, the Status Toggle function can be enabled or disabled.

Bit 13 can be used to extend the Status Toggle information to a 2-Bit Input Cycle Counter. Object $60D9_h$ and $60DA_h$ shall be supported and used to enable or disable the Input Cycle Counter functionality.

The Bit 12 *Drive follows the command value* shall be zero if the drive **does not** follow the target value (position, velocity or torque) because of local reasons (internal set-point settings), e.g. if a local Input is configured to a halt function or if a safety function prevents the drive in Operational to follow the target set point. The control device shall evaluate the bit. The Bit 12 shall be set if the drive is in state *Operation enabled* and follows the target and set-point values of the control device. In all other cases it shall be zero. If the bit is not supported it shall be fix set to 1 in the statusword.

NOTE: If Bit 12 is not supported and is not set to 1, the control device will not run the drive, because it expects that the drive does not follow the command.

6.4 Cyclic synchronous torque mode (cst)

Figure 9 shows an overview of the *Cyclic synchronous torque mode (cst)*.

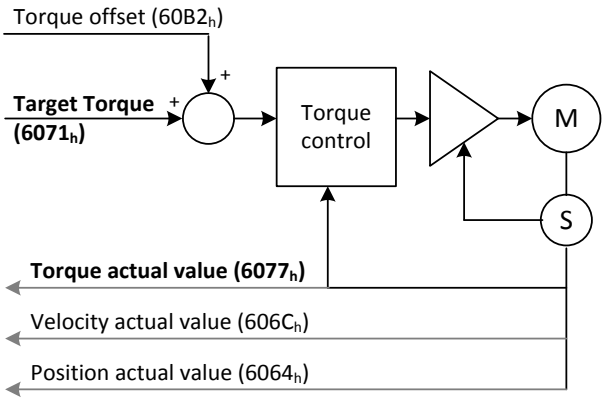


Figure 9: Cyclic synchronous torque mode overview

The following (bold) objects are used for *cst* (Figure 10).

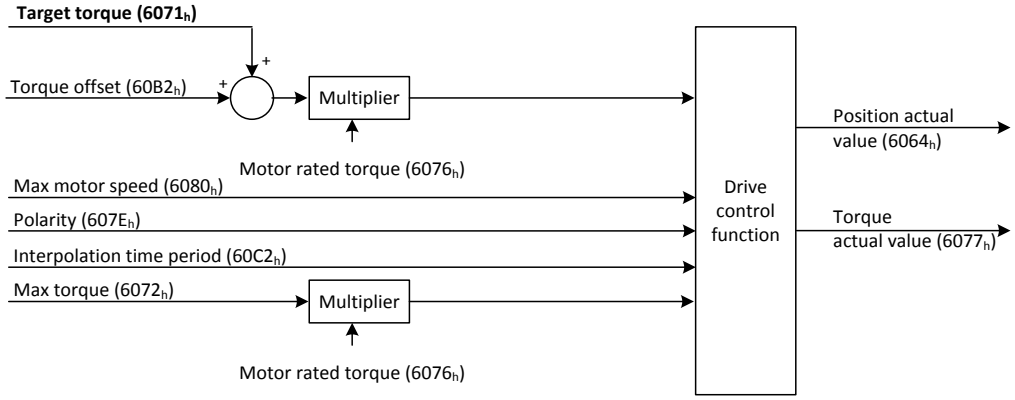


Figure 10: Cyclic synchronous torque control function

Table 11: cst mode – Object list

Index	Name	Category
6071h	**Target Torque**	**M**
6077h	**Torque actual value**	**M**
6064h	Position actual value	R
60C2h	Interpolation time period	R

In the cst mode the Halt bit (bit 8) of the Controlword shall be ignored because the halt function is controlled by the control device.

Bit 5 and 6 of Controlword can be used as Output Cycle Counter. This 2-Bit counter can be used by the control device to indicate if updated output data are available. The counter shall be incremented with every update of the output process data. Object 60D9h and 60DAh shall be supported and used to enable or disable the Output Cycle Counter functionality.

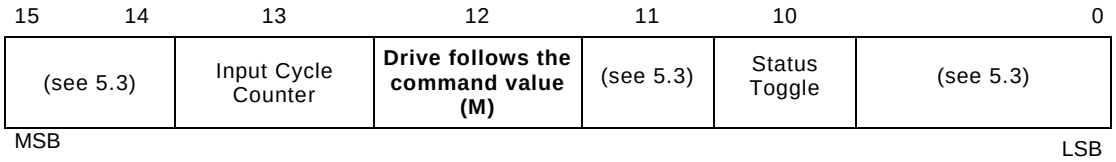


Figure 11: Statusword for *cst* mode

In the Statusword Bit 12 is mandatory.

In *cst* mode Bit 10 can be used as *Status Toggle* information to indicate if the device provides updated input data. The bit shall be toggled with every update of the input process data. If object $60D9_h$ is supported, the Status Toggle function can be enabled or disabled.

Bit 13 can be used to extend the Status Toggle information to a 2-Bit Input Cycle Counter. Object $60D9_h$ and $60DA_h$ shall be supported and used to enable or disable the Input Cycle Counter functionality.

The Bit 12 *Drive follows the command value* shall be zero if the drive **does not** follow the target value (position, velocity or torque) because of local reasons (internal set-point settings), e.g. if a local Input is configured to a halt function or if a safety function prevents the drive in Operational to follow the target set point. The control device shall evaluate the bit. The Bit 12 shall be set if the drive is in state *Operation enabled* and follows the target and set-point values of the control device. In all other cases it shall be zero. If the bit is not supported it shall be fix set to 1 in the statusword.

NOTE: If Bit 12 is not supported and is not set to 1, the control device will not run the drive, because it expects that the drive does not follow the command.

6.5 Cyclic synchronous torque mode with commutation angle (cstca)

6.5.1 General information

The overall structure for this mode is shown in Figure 12. With this mode, the trajectory generator is located in the control device, not in the drive device. In cyclic synchronous manner, it provides a commutation angle and a target torque to the drive device, which performs current control and space vector modulation. Optionally, an additive torque value can be provided by the control system in order to allow two instances to set up the torque. Measured by sensors, the drive device could provide actual values for position or may provide velocity and torque to the control device.

This mode can be used for example

- to find the commutation angle during commissioning
- to check the function (increments, zero signal) and direction of the sensor
- Use of external sensor interface to calculate commutation angle (in that case there might be no internal feedback to Torque control)

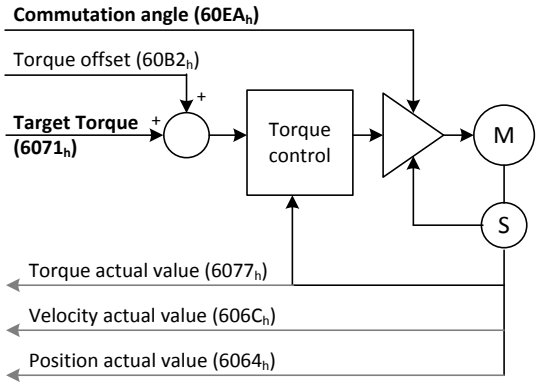


Figure 12: Cyclic synchronous torque mode with commutation angle overview

6.5.2 General definitions

The factors necessary for scaling have a linear relationship and therefore they are described in the factor group. The polarity is described in the factor group as well.

6.5.3 Functional description

Figure 13 shows the inputs and outputs of the torque control function with commutation angle. The inputs (from the control function point of view) are the commutation angle and the target torque and optionally a torque offset (to be added to the target torque to allow two instances to set up the torque).

The drive device can have features for limitation of motor speed. The torque can be limited as well.

The interpolation time period defines the time period between two updates of the target velocity and/or additive velocity and shall be used for inter-cycle interpolation.

The torque actual value can be used as output to the control device.

The following (bold) objects are used for *cstca* (Figure 13).

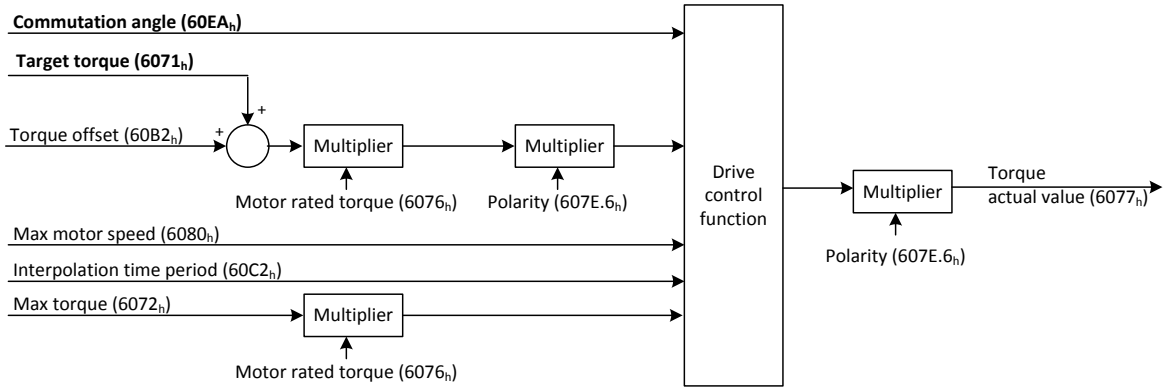


Figure 13: Cyclic synchronous torque with commutation angle control function

Table 12: cstca mode – Object list

Index	Name	Category
6071$_h$	**Target Torque**	**M**
60EA$_h$	**Commutation angle**	**M**
6077$_h$	Torque actual value	R
6064$_h$	Position actual value	R

In the *cstca* mode the Halt bit (bit 8) of the Controlword shall be ignored because the halt function is controlled by the control device.

Bit 5 and 6 of Controlword can be used as Output Cycle Counter. This 2-Bit counter can be used by the control device to indicate if updated output data are available. The counter shall be incremented with every update of the output process data. Object 60D9$_h$ and 60DA$_h$ shall be supported and used to enable or disable the Output Cycle Counter functionality.

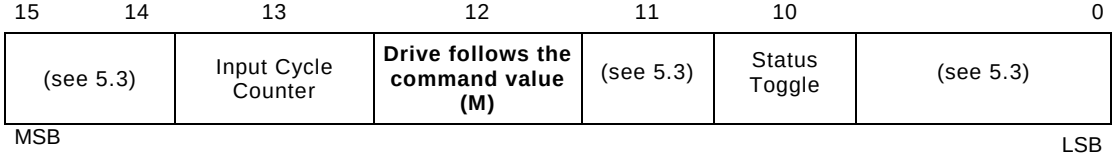


Figure 14: Statusword for *cstca* mode

In the Statusword Bit 12 is mandatory.

In *cstca* mode Bit 10 can be used as *Status Toggle* information to indicate if the device provides updated input data. The bit shall be toggled with every update of the input process data. If object 60D9$_h$ is supported, the Status Toggle function can be enabled or disabled.

Bit 13 can be used to extend the Status Toggle information to a 2-Bit Input Cycle Counter. Object $60D9_h$ and $60DA_h$ shall be supported and used to enable or disable the Input Cycle Counter functionality.

The Bit 12 *Drive follows the command value* shall be zero if the drive **does not** follow the target value (position, velocity or torque) because of local reasons (internal set-point settings), e.g. if a local Input is configured to a halt function or if a safety function prevents the drive in Operational to follow the target set point. The control device shall evaluate the bit. The Bit 12 shall be set if the drive is in state *Operation enabled* and follows the target and set-point values of the control device. In all other cases it shall be zero. If the bit is not supported it shall be fix set to 1 in the statusword.

NOTE: If Bit 12 is not supported and is not set to 1, the control device will not run the drive, because it expects that the drive does not follow the command.

6.5.4 Object $60EA_h$: Commutation angle

This object shall indicate the electrical commutation angle for the space vector modulation. The value shall be given in $360°/2^{16}$, whereby the electrical angle is used. Table 13 specifies the object description, and Table 14 specifies the entry description.

Table 13: Object description

Attribute	Value
Index	$60EA_h$
Name	Commutation angle
Object Code	Variable
Data Type	Unsigned16
Category	Mandatory if *cstca* is supported

Table 14: Entry description

Attribute	Value
Sub-Index	00_h
Access	rw
PDO Mapping	optional
Value Range	Unsigned16
Default Value	0000_h

6.6 Profile Position mode (pp)

For information about the profile position mode refer to IEC 61800-7-201, [3]. Table 15 shows the object list of pp mode

Table 15: pp mode – Object list

Index	Name	Category
607A$_h$	**Target Position**	**M**
6064$_h$	**Position actual value**	**M**
6081$_h$	**Profile velocity**	**M**
6083$_h$	**Profile acceleration**	**M**

The pp mode is part of this directive since it is widely used in drive implementations. Support of pp mode is recommended for drives that are handled mainly by a PLC based control unit

6.7 Polarity object

The polarity object information (607E$_h$) should be used for all axes-specific value. I.e. for

- Position actual value (6064$_h$)
- Following error actual value (60F4$_h$)
- Velocity actual value (606C$_h$)
- Torque actual value (6077$_h$)

The Velocity sensor actual value (6069$_h$) shall reflect the original values of the sensor.

It is highly recommended that position polarity (607E$_h$.7) and velocity polarity (607E$_h$.6) have the same value. It can be different, if two devices are used – one for position and another one for velocity.

NOTE: Polarity for Torque is taken from either position polarity (607E.7) or velocity polarity (607E.6) . It depends on the application

6.8 Object definitions

6.8.1 Object 6502$_h$: Supported drive modes

This object shall provide information on the supported drive modes. Figure 15 specifies the value definition, Table 16 specifies the object description, and Table 17 specifies the entry description.

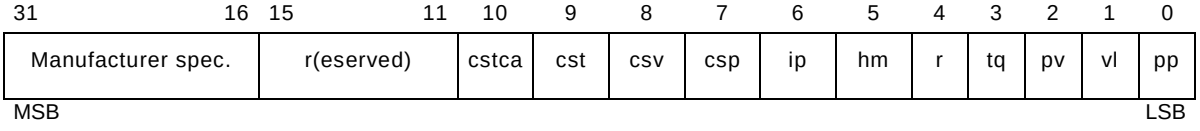


Figure 15: Value definition

Table 16: Object description

Attribute	Value
Index	6502$_h$
Name	Supported drive modes
Object Code	Variable
Data Type	Unsigned32
Category	Mandatory

Table 17: Entry description

Attribute	Value
Sub-Index	00$_h$
Access	ro
PDO Mapping	possible

Attribute	Value
Value Range	see Figure 15
Default Value	No

6.8.2 Object 60D9$_h$: Supported synchronization functions

This object shall provide information on the supported synchronization functions in the device.

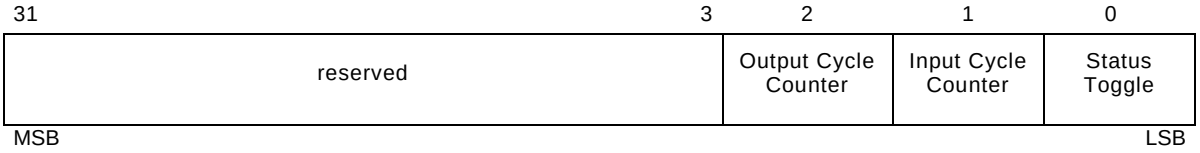


Figure 16: Value definition

1 = Function supported, 0 = Function not supported
Status Toggle: Status Toggle bit in csp, csv, cst and cstca mode supported
Input Cycle Counter: Status Toggle can be extended to a 2-Bit Input Cycle Counter in the Status word in csp, csv, cst and cstca mode
Output Cycle Counter: Output Cycle Counter in csp, csv, cst and cstca mode supported

Figure 16 specifies the value definition, Figure 15 specifies the object description, and Figure 16 specifies the entry description.

Table 18: Object description

Attribute	Value
Index	60D9$_h$
Name	Supported functions
Object Code	Variable
Data Type	Unsigned32
Category	Optional

Table 19: Entry description

Attribute	Value
Sub-Index	00$_h$
Access	Ro
PDO Mapping	No
Value Range	see Figure 16
Default Value	manufacturer specific

6.8.3 Object 60DA$_h$: Synchronization function settings

This object shall enable/disable supported synchronization functions in the device.

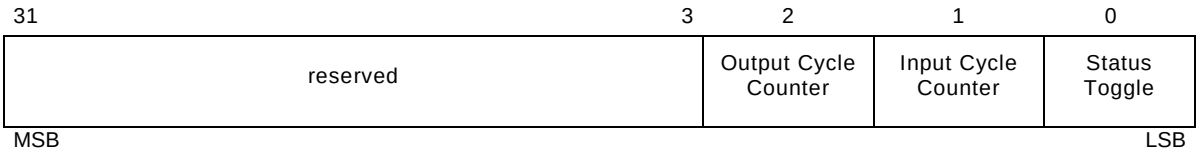


Figure 17: Value definition

Status Toggle bit[1]:
1 = Status Toggle in status word in csp, csv, cst and cstca mode enabled,
0 = Status Toggle in status word in csp, csv, cst and cstca mode disabled
Input Cycle Counter bit[1]:
1 = Input Cycle counter in status word in csp, csv, cst and cstca mode enabled,
0 = Input Cycle counter in status word in csp, csv, cst and cstca mode disabled
Output Cycle Counter bit:
1 = Output Cycle Counter in control word in csp, csv, cst and cstca mode enabled,
0 = Output Cycle Counter in control word in csp, csv, cst and cstca mode disabled

If Output Cycle Counter is enabled and the drive detects an error it shall behave as described in ETG.1020 Error Handling for RxPDO Toggle, see [9].

Status Toggle resp. Input Cycle Counter are independent of Output Cycle Counter.

Figure 17 specifies the value definition, Table 20 specifies the object description, and Table 21 specifies the entry description.

Table 20: Object description

Attribute	Value
Index	60DA$_h$
Name	Function settings
Object Code	Variable
Data Type	Unsigned32
Category	Optional

Table 21: Entry description

Attribute	Value
Sub-Index	00$_h$
Access	rw
PDO Mapping	no
Value Range	see Figure 17
Default Value	0000 0000$_h$

The control unit can write the requested mode to 60DA$_h$ during start-up. Following reactions from the drive shall be accepted:

a) Drive accepts values
The control unit and the drive shall use the requested synchronization function

b) If none of the synchronization functions are supported and therefore the object does not exist the drive sends abort with Abort Code 06020000$_h$ “The object does not exist in the object directory”.
The control unit shall not use the synchronization function

c) If the requested mode is not supported the drive sends abort with Abort Code 06090030$_h$ “Value range of parameter exceeded”.

[1] If Status Toggle and Input Cycle Counter are both enabled, the Input Cycle Counter functionality shall be used.

The control unit shall not use the requested synchronization function.
The drive shall not use any synchronization function, i.e. $60DA_h = 0000_h$

6.9 Switching between Operation modes

The dynamic change of the operation mode in state operation is defined in this chapter. The Operation mode can be switched by writing the Object 6060_h Modes of operation. This can be done via PDO communication or via SDO communication

6.9.1 Static setting of operation mode

Cyclic PDO communication and acyclic SDO communication are often implemented in different tasks, so there is no synchronization between an object access (e.g. SDO write-access to object “mode of operation”) and the cyclic process data processing in the drives. If the control device selects another operation mode via SDO, there might be the critical behavior, that the update of the “new selected” process data objects is too late, so the drive works with older or undefined process data values (e.g. wrong target position) until a new PDO is received.

The selection of the operation mode via SDO (or local device configuration tool) can be done in state “pre-op”.

For the device parameterization the Index 6060_h Modes of operation shall be part of the Start-up parameter. The list of possible operation modes can be described in the device description file (ESI) in form of several devices or in form of different modules with different start-up parameters. The user can select the operation mode in the configuration tool.

EXAMPLE:

A Drive supports the csp mode and the csv mode. In the ESI there are two modules defined:

Module A for csp mode: This module contains the PDO mapping for the csp mode (RxPDO: Controlword, Target Position; TxPDO: Statusword, Position actual Value). It contains the start-up parameter 06060_h = 8 (csp)

Module B for csv mode: This module contains the PDO mapping for csv mode (RxPDO: Controlword, Target Velocity; TxPDO: Statusword, Velocity actual Value). It contains the start-up parameter 6060_h = 9 (csv)

6.9.2 Dynamic switching of operation mode in state operation

For a dynamic switching of the operation mode in state operation, it is recommended to select the mode of operation via cyclic PDO communication.

The object 6060_h Modes of operation shall be mapped in the RxPDO and the object 6061_h Modes of operation display in the TxPDO. Additional the mode specific objects that are required for the dynamically selectable modes of operation shall be mapped in the Rx/TxPDO, too.

The control device has the responsibility to update all operation mode specific process data objects together with the selection of the operation mode at the same time. In this case there is no critical runtime behavior between the operation mode selection and the content of the process data objects.

If the control device selects a new operation mode, it depends on the drive implementation, if the drive switches during one process data cycle to the new operation mode. Mostly, the drives will answer with the process data information of the “old” operation mode, until the internal switching to the new operation mode is done. For the intermediate time, the control device shall transmit valid data in the PDO for the old and the new operation mode.

The mode specific bits in the Controlword shall always fit to the "Mode of operation" in the RxPDO, even in the intermediate time.

The mode specific bits in the Statusword shall always fit to the "Modes of operation display" in the TxPDO, even in the intermediate time.

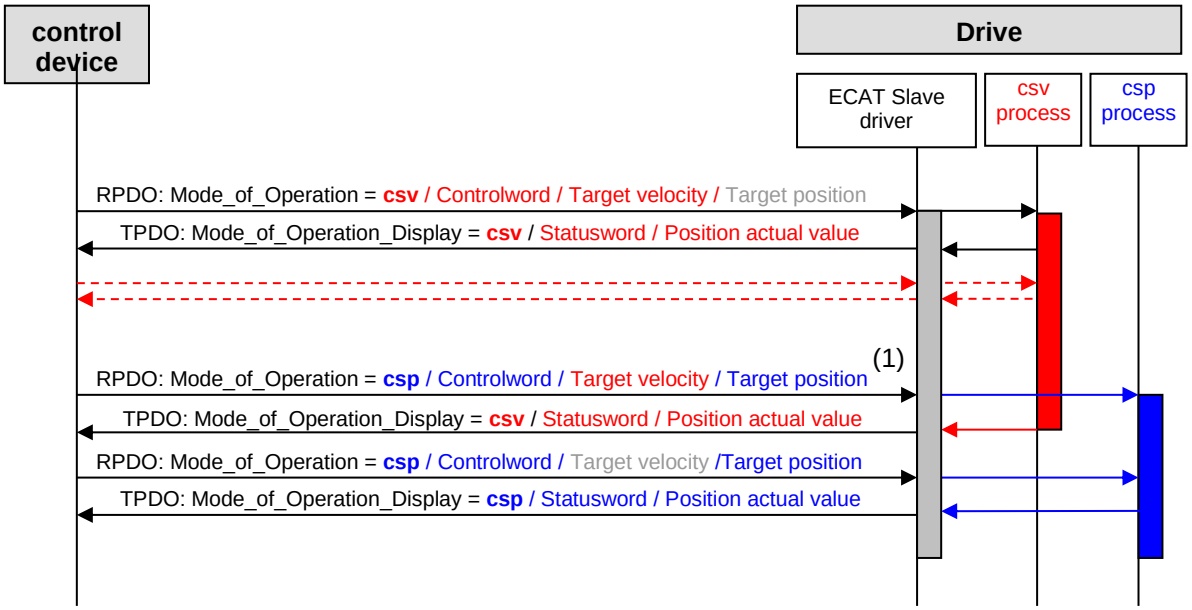


Figure 18: Dynamic change of operation mode

7 Function Groups

7.1 General

To describe the behavior of a servo drive, different function groups (FG) are defined in this directive. The drive does not have to support all of these function groups, but if a group is supported it has to fulfill the specified objects and features.

The mandatory objects and features in the FGs are necessary for functionality and quality of the specified function.

7.2 Function Group "Torque Limiting"

The Function Group "Torque Limiting" defines

Table 22: FG Torque Limiting – Object list

Index	Name	Category	Remark
6072_h	Max torque	**M**	Mandatory if FG Torque Limiting is supported
$60E0_h$	Positive torque limit value	**C**	Mandatory if FG Torque Limiting is supported
$60E1_h$	Negative torque limit value	**C**	Mandatory if FG Torque Limiting is supported

The limiting of the torque will be stated in the bit 11 "internal limit active" in the statusword.

The Figure 19 shows the usage of the three limiting values

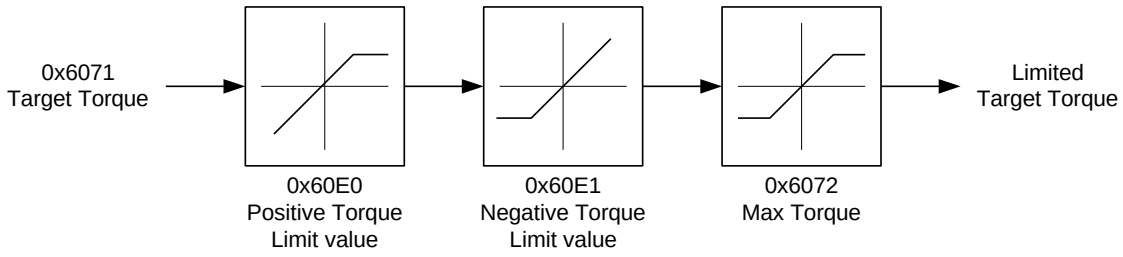


Figure 19: Usage of torque limiting objects

The lowest of the limiting values is effective.

7.2.1 Object $60E0_h$: Positive torque limit value

This object shall indicate the configured maximum positive torque in the motor. The value shall be given per thousand of rated torque. Table 23 specifies the object description, and Table 24 specifies the entry description.

Table 23: Object description object $60E0_h$

Attribute	Value
Index	$60E0_h$
Name	Positive Torque Limit
Object Code	Variable
Data Type	Unsigned16
Category	Mandatory if FG "Torque Limiting" is supported

Table 24: Entry description object $60E0_h$

Attribute	Value
Sub-Index	00_h
Access	rw

Attribute	Value
PDO Mapping	possible
Value Range	Unsigned16
Default Value	Manufacturer-specific

Positive torque takes effect in the case of:

- motive operation is positive velocity
- regenerative operation is negative velocity

7.2.2 Object $60E1_h$: Negative torque limit value

This object shall indicate the configured maximum negative torque in the motor. The value shall be given per thousand of rated torque. Table 25specifies the object description, and Table 26 specifies the entry description.

Table 25: Object description object $60E1_h$

Attribute	Value
Index	$60E1_h$
Name	Negative Torque Limit
Object Code	Variable
Data Type	Unsigned16
Category	Mandatory if FG "Torque Limiting" is supported

Table 26: Entry description object $60E1_h$

Attribute	Value
Sub-Index	00_h
Access	rw
PDO Mapping	possible
Value Range	Unsigned16
Default Value	Manufacturer-specific

Negative torque takes effect in the case of:

- motive operation is negative velocity
- regenerative operation is positive velocity

7.3 Homing

The homing is generally leaded by the control device. Therefore all homing methods are optional.

Table 27: Homing methods

Method	Description	Category
1	Homing on negative limit switch and index pulse	O
2	Homing on positive limit switch and index pulse	O
3, 4	Homing on positive home switch and index pulse	O
5, 6	Homing on negative home switch and index pulse	O
7 …14	Homing on home switch and index pulse	O
15, 16	Reserved	O
17 …30	Homing without index pulse	O
31, 32	Reserved	O
33, 34	Homing on index pulse	O
35	Homing on current position – obsolete	--
~~36~~	~~Homing with touch-probe~~	~~R~~
37	Homing on current position	O

7.3.1 Homing Method 35: Homing on current position (obsolete)

In this method, the current position shall be taken to be the home position. This method does not require the drive device to be in operational enabled state.

NOTE This method is obsolete because of compatibility reasons.

7.3.2 Homing Method 36: Homing with touch-probe – reserved for compatibility reasons

This method was **not** used as a Homing Mode.

The control device leads the drive, e.g. in csp, csv or cst operation mode. For a high precise accuracy, the control device activates the touch probe inputs of the drive and use the touch probe information for detection of the homing position.

NOTE: To define this method as obsolete/reserved was decided by CiA SIG. Use Homing Method 37 instead

7.3.3 Homing Method 37: Homing on current position

In this method, the current position shall be taken to be the home position. This method does not require the drive device to be in operational enabled state.

At the Home position (i.e. after the homing process) the position actual value (6064_h) is calculated as follows:

Position actual value (6064_h) = Home Offset ($607C_h$)

EXAMPLE: Applications with no absolute sensor information.

EXAMPLE: The control device can set the operation mode csp, csv or cst and runs the drive to the home position. At the home position the control device switches the operation mode to Homing mode, Method 37. The drive takes the current position as the home position. Then the control device switches back to operation mode csp, csv or cst.

7.3.4 Calculation of position actual value by homing process

The current position is the unprocessed position sensor information (incremental, single or multi turn sensor) before homing.

For a single turn sensor the single turn information represents the position sensor information. For a multi turn sensor the multi turn information represents the position sensor information. For an incremental sensor (e.g. TTL encoder) the position sensor information is zero after initialization.

The Home Offset is configured with the machine specific value during commissioning.

During homing, the machine home position is found and once the homing is completed, the zero position is offset from the home position by adding the home offset to the home position.

Zero position = home position + home offset

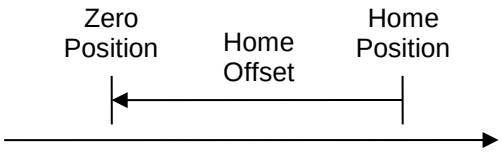


Figure 20: Home Offset definition

The configured Home Offset ($607C_h$) is always used for the calculation of the Position actual value (6064_h) during Homing process!

The position actual value (6064_h) is the current software position in the drive. It is based on the unprocessed position sensor information and the Home Offset (after Homing process).

NOTE: The activation of a new value of the object home offset is manufacturer-specific. It is recommended to apply the new value only while the drive is in homing mode.

At the Home position (i.e. after the homing process) the position actual value (6064_h) is calculated according to the used Homing method.

7.3.5 Homing object list

Table 28: Homing – Object list

Index	Name	Category	Remark
$607C_h$	Home Offset	R	Recommended if Homing is supported
6098_h	Homing method	**M**	Mandatory if Homing is supported
6099_h	Homing speed	O	
$609A_h$	Homing acceleration	O	
$60E3_h$	Supported homing methods	R	Recommended if Homing is supported

In the controlword the Bit 4 is used.

15…9	9	8…7	6	5	4	3…0
(see 5.2)	reserved	(see 5.2)	reserved	**Homing operation start (M)**		(see 5.2)
MSB						LSB

7.3.6 Object $60E3_h$: Supported Homing Methods

This object defines the supported homing methods of the drive. Table 29 specifies the object description, and Table 30 specifies the entry description.

Table 29: Object description object $60E3_h$

Attribute	Value
Index	$60E3_h$
Name	Supported Homing Methods
Object Code	ARRAY
Data Type	Integer8
Category	Conditional, Recommended if FG Homing is supported

NOTE: Only for legacy reasons data type Integer16 is allowed for this object

Table 30: Entry description object $60E3_h$

Attribute	Value
Sub-Index	00_h
Description	Highest sub-index supported
Entry category	Mandatory
Access	ro
PDO Mapping	no
Value Range	1 to 254
Default Value	Manufacturer-specific
Sub-Index	01_h
Description	1st supported homing method
Entry category	Mandatory
Access	ro
PDO Mapping	no
Value Range	see Figure 21
Default Value	Manufacturer-specific
Sub-Index	02_h
Description	2nd supported homing method
Entry category	Optional
Access	ro
PDO Mapping	no
Value Range	see Figure 21
Default Value	Manufacturer-specific
to	
Sub-Index	FE_h
Description	254th supported homing method
Entry category	Optional
Access	ro
PDO Mapping	no
Value Range	see Figure 21
Default Value	Manufacturer-specific

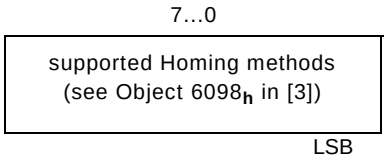


Figure 21: Object $60E3_h$ Supported Homing Methods – Value definition

7.4 Function Group "Touch Probe"

For the Function Group "Touch Probe" at minimum one local digital input at the servo drive shall have a touch probe capability. The following objects are mandatory.

Table 31: FG Touch probe – Object list

Index	Name	Category
$60B8_h$	**Touch probe function**	**M**
$60B9_h$	**Touch probe status**	**M**
$60BA_h$	**Touch probe position 1 positive value**	**M**
$60BB_h$	**Touch probe position 1 negative value**	**M**
$60D0_h$	Touch probe source	R
$60D1_h$	Touch probe time stamp 1 positive value	O
$60D2_h$	Touch probe time stamp 1 negative value	O
$60D5_h$	Touch probe 1 positive edge counter	O
$60D6_h$	Touch probe 1 negative edge counter	O

If a second Touch Probe input is supported the following objects are used

Table 32: Touch Probe objects for optional Input 2

Index	Name	Category
$60BC_h$	Touch probe position 2 positive value	O
$60BD_h$	Touch probe position 2 negative value	O
$60D3_h$	Touch probe time stamp 2 positive value	O
$60D4_h$	Touch probe time stamp 2 negative value	O
$60D7_h$	Touch probe 2 positive edge counter	O
$60D8_h$	Touch probe 2 negative edge counter	O

The source of the touch probe inputs can differ. E.g. it could be a digital input or a signal from the encoder or something else. This is defined manufacturer specific.

Optional more than two touch probe inputs can be implemented in the drive. The manufacturer specific objects used for these additional inputs shall correspond to the standard touch probe objects.

7.4.1 Object 60B8 $_h$: Touch probe function

This object shall indicate the configured function of the touch probe. Table 33 specifies the value definition.

The definition of bit 3 is extended according to the IEC 61800-7 to configure additional touch probe sources.

Table 33: Value definition object 60B8$_h$

Bit	Value	Definition
0	0	Switch off touch probe 1
	1	Enable touch probe 1
1	0	Trigger first event
	1	Continuous
3,2	00	Trigger with touch probe 1 input,
	01	Trigger with zero impulse signal or position encoder,
	10	Touch probe source defined by object 60D0$_h$.1
	11	Reserved
4	0	Switch off sampling at positive edge of touch probe 1
	1	Enable sampling at positive edge of touch probe 1
5	0	Switch off sampling at negative edge of touch probe 1
	1	Enable sampling at negative edge of touch probe 1
6, 7	-	User defined (e.g. for testing)
8	0	Switch off touch probe 2
	1	Enable touch probe 2
9	0	Trigger first event
	1	Continuous
11,10	00	Trigger with touch probe 2 input
	01	Trigger with zero impulse signal or position encoder
	10	Touch probe source defined by object 60D0$_h$.2
	11	reserved
12	0	Switch off sampling at positive edge of touch probe 2
	1	Enable sampling at positive edge of touch probe 2
13	0	Switch off sampling at negative edge of touch probe 2
	1	Enable sampling at negative edge of touch probe 2
14, 15	-	User defined (e.g. for testing)

7.4.2 Object 60D0$_h$: Touch probe source

This object defines the source of the touch probe functions. Table 34 lists the value definition of the object.

Table 34: Value definition

Value	Definition
-32 768 … -1	Manufacturer-specific
0	reserved
+1	Digital Input 1 (Touch probe input)
+2	Digital Input 2 (Touch probe input)
+3	Digital Input 3 (Touch probe input)

Value	Definition
+4	Digital Input 4 (Touch probe input)
+5	Hardware Zero impulse signal of position encoder
+6	Software Zero impulse signal of position encoder
+7 … +32 767	reserved

Table 35 specifies the object description, and Table 36 specifies the entry description.

Table 35: Object description

Attribute	Value
Index	$60D0_h$
Name	Touch probe source
Object Code	ARRAY
Data Type	Integer16
Category	Conditional, Recommended if FG Touch probe is supported

Table 36: Entry description

Attribute	Value
Sub-Index	00_h
Description	Highest sub-index supported
Entry category	Mandatory
Access	ro
PDO Mapping	no
Value Range	1 to 254
Default Value	Manufacturer-specific
Sub-Index	01_h
Description	Touch probe 1 source
Entry category	Mandatory
Access	rw
PDO Mapping	no
Value Range	see Table 34
Default Value	Manufacturer-specific
Sub-Index	02_h
Description	Touch probe 2 source
Entry category	Optional
Access	rw
PDO Mapping	no
Value Range	see Table 34
Default Value	Manufacturer-specific
	to
Sub-Index	FE_h
Description	Touch probe 254 source
Entry category	Optional
Access	rw

Attribute	Value
PDO Mapping	no
Value Range	see Table 34
Default Value	Manufacturer-specific

7.4.3 Object 60B9$_h$: Touch probe status

This object shall provide the status of the touch probe. Table 37 specifies the value definition.

Table 37: Value definition object 60B9$_h$

Bit	Value	Definition
0	0	Touch probe 1 is switched off
	1	Touch probe 1 is enabled
1	0	Touch probe 1 no positive edge value stored
	1	Touch probe 1 positive edge position stored
2	0	Touch probe 1 no negative edge value stored
	1	Touch probe 1 negative edge position stored
3 – 5	0	Reserved
6, 7	-	User defined (e.g. for testing) NOTE: if not otherwise used, it is recommended to use bit 7 to show the current state of Touch probe 1 input signal
8	0	Touch probe 2 is switched off
	1	Touch probe 2 is enabled
9	0	Touch probe 2 no positive edge value stored
	1	Touch probe 2 positive edge position stored
10	0	Touch probe 2 no negative edge value stored
	1	Touch probe 2 negative edge position stored
11 – 13	0	Reserved
14, 15	-	User defined (e.g. for testing) NOTE: if not otherwise used, it is recommended to use bit 15 to show the current state of Touch probe 2 input signal

The Touch probe status information shall be evaluated first by the control device, to check if the stored Touch probe values (60BA$_h$ ... 60BD$_h$) are valid.

NOTE: Bit 1 and bit 2 are set to 0$_b$ when touch probe 1 is switched off (object 60B8$_h$ bit 0 is 0$_b$). Bit 9 and 10 are set to 0$_b$ when touch probe 2 is switched off (object 60B8$_h$ bit 8 is 0$_b$).

Figure 22 shows a timing diagram for a touch probe configuration and the corresponding behavior.

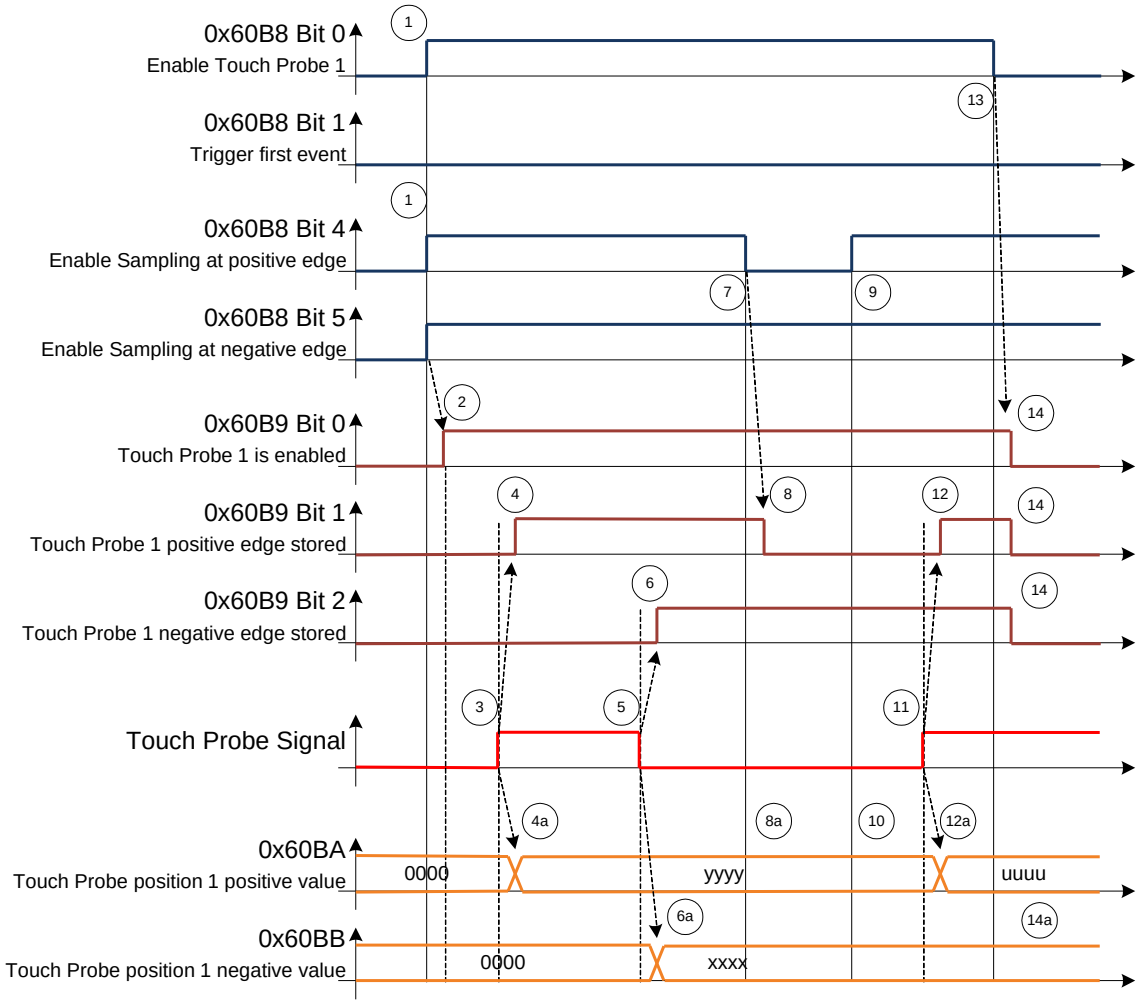


Figure 22: Timing diagram for Touch probe example

Table 38: Touch Probe Timing example

Number	Touch probe behavior	
(1)	$60B8_h$, Bit 0 = 1 $60B8_h$ Bit 1, 4, 5	Enable Touch Probe 1, Configure and Enable Touch Probe 1 positive and negative edge
(2)	→ $60B9_h$ Bit 0 = 1	Status “Touch Probe 1 enabled” is set
(3)	External touch probe signal has positive edge	
(4)	→ $60B9_h$ Bit 1 = 1	Status “Touch Probe 1 positive edge stored” is set
(4a)	→ $60BA_h$	Touch probe position 1 positive value is stored
(5)	External touch probe signal has negative edge	
(6)	→ $60B9_h$ Bit 2 = 1	Status “Touch Probe 1 negative edge stored” is set
(6a)	→ $60BB_h$	Touch probe position 1 negative value is stored
(7)	$60B8_h$:4	Sample positive edge is disabled
(8)	→ $60B9_h$ Bit 0 = 0	Status “Touch Probe 1 positive edge stored” is reset
(8a)	→ $60BA_h$	Touch probe position 1 positive value is not changed
(9)	$60B8_h$ Bit 4 = 1	Sample positive edge is enabled
(10)	→ $60BA_h$	Touch probe position 1 positive value is not changed
(11)	External touch probe signal has positive edge	
(12)	→ $60B9_h$ Bit 1 = 1	Status “Touch Probe 1 positive edge stored” is set
(12a)	→ $60BA_h$	Touch probe position 1 positive value is stored
(13)	$60B8_h$ Bit 0 = 0	Touch Probe 1 is disabled
(14)	→ $60B9_h$ Bit 0, 1, 2 = 0	Status Bits are reset
(14a)	→ $60BA_h$, $60BB_h$	Touch probe position 1 positive/negative value are not changed

7.4.4 Touch probe time stamp latch

In addition to the position ($60BA_h$, $60BB_h$, $60BC_h$, $60BD_h$) the current time stamp can be latched by the device.

The objects $60D1_h$, $60D2_h$, $60D3_h$, $60D4_h$ should capture the corresponding time stamp for Touch probe 1 and 2.

NOTE: If the device supports Distributed Clocks (DC) and the DC is activated, the lower 32 Bits of the DC Unit shall be used as time stamp values

The control device can use several process values with the same time stamp to calculate new set points.

7.4.4.1 Object $60D1_h$: Touch probe time stamp 1 positive value

This object shall provide the time stamp value of the touch probe 1 at positive edge. The value shall be given in nanoseconds. Table 39 specifies the object description, and Table 40 specifies the entry description.

Table 39: Object description

Attribute	Value
Index	$60D1_h$
Name	Touch probe time stamp 1 positive value
Object Code	VAR
Data Type	Unsigned32
Category	Optional

Table 40: Entry description

Attribute	Value
Sub-Index	00_h
Access	ro
PDO Mapping	TxPDO
Value Range	Unsigned 32
Default Value	no

7.4.4.2 Object $60D2_h$: Touch probe time stamp 1 negative value

This object shall provide the time stamp value of the touch probe 1 at negative edge. The value shall be given in nanoseconds. Table 41 specifies the object description, and Table 42 specifies the entry description.

Table 41: Object description

Attribute	Value
Index	$60D2_h$
Name	Touch probe time stamp 1 negative value
Object Code	VAR
Data Type	Unsigned32
Category	Optional

Table 42: Entry description

Attribute	Value
Sub-Index	00_h
Access	ro
PDO Mapping	TxPDO
Value Range	Unsigned 32

Attribute	Value
Default Value	no

7.4.4.3 Object 60D3$_h$: Touch probe time stamp 2 positive value

This object shall provide the time stamp value of the touch probe 2 at positive edge. The value shall be given nanoseconds. Table 43 specifies the object description, and Table 44 specifies the entry description.

Table 43: Object description

Attribute	Value
Index	60D3$_h$
Name	Touch probe time stamp 2 positive value
Object Code	VAR
Data Type	Unsigned32
Category	Optional

Table 44: Entry description

Attribute	Value
Sub-Index	00$_h$
Access	ro
PDO Mapping	TxPDO
Value Range	Unsigned 32
Default Value	no

7.4.4.4 Object 60D4$_h$: Touch probe time stamp 2 negative value

This object shall provide the time stamp value of the touch probe 2 at negative edge. The value shall be given nanoseconds. Table 45 specifies the object description, and Table 46 specifies the entry description.

Table 45: Object description

Attribute	Value
Index	60D4$_h$
Name	Touch probe time stamp 2 negative value
Object Code	VAR
Data Type	Unsigned32
Category	Optional

Table 46: Entry description

Attribute	Value
Sub-Index	00$_h$
Access	ro
PDO Mapping	TxPDO
Value Range	Unsigned 32
Default Value	no

7.4.5 Touch probe edge counter for continuous mode

For continuous touch probe mode ($60B8_h$ Bit 1 = 1 or $60B8_h$ Bit 9 = 1) a counter per touch probe channel is incremented with each touch probe event. The control unit can check how many touch probe events happens between the control cycles.

Per touch probe and per edge a counter object is defined.

$60D5_h$ Touch probe 1 positive edge counter
$60D6_h$ Touch probe 1 negative edge counter
$60D7_h$ Touch probe 2 positive edge counter
$60D8_h$ Touch probe 2 negative edge counter

NOTE The counters are continuous counters. Counter overrun has to be taken into account

The according Touch probe counter is only valid if Touch probe input is enabled ($60B8_h$: Bit 0 / Bit 8). The value shall be interpreted depending on the mode:

- Single event measuring
  Only Bit 0 is valid,
- Continuous measuring
  This is a 16 Bit value with overflow

7.4.5.1 Object $60D5_h$: Touch probe 1 positive edge counter

This object shall provide a continuous counter that is incremented with each positive edge at touch probe 1. The counter is only valid if Touch probe input is enabled ($60B8_h$: Bit 0 = 1). Table 47 specifies the object description, and Table 48 specifies the entry description.

Table 47: Object description

Attribute	Value
Index	$60D5_h$
Name	Touch probe 1 positive edge counter
Object Code	VAR
Data Type	Unsigned16
Category	Optional

Table 48: Entry description

Attribute	Value
Sub-Index	00_h
Access	Ro
PDO Mapping	TxPDO
Value Range	Unsigned 16
Default Value	No

7.4.5.2 Object $60D6_h$: Touch probe 1 negative edge counter

This object shall provide a continuous counter that is incremented with each negative edge at touch probe 1. The counter is only valid if Touch probe input is enabled ($60B8_h$: Bit 0 = 1). Table 70 specifies the object description, and Table 71 specifies the entry description.

Table 49: Object description

Attribute	Value
Index	$60D6_h$
Name	Touch probe 1 negative edge counter
Object Code	VAR
Data Type	Unsigned16

Attribute	Value
Category	Optional

Table 50: Entry description

Attribute	Value
Sub-Index	00_h
Access	ro
PDO Mapping	TxPDO
Value Range	Unsigned 16
Default Value	no

7.4.5.3 Object $60D7_h$: Touch probe 2 positive edge counter

This object shall provide a continuous counter that is incremented with each positive edge at touch probe 2. The counter is only valid if Touch probe input is enabled ($60B8_h$: Bit 8 = 1). Table 51 specifies the object description, and Table 52 specifies the entry description.

Table 51: Object description

Attribute	Value
Index	$60D7_h$
Name	Touch probe 2 positive edge counter
Object Code	VAR
Data Type	Unsigned16
Category	Optional

Table 52: Entry description

Attribute	Value
Sub-Index	00_h
Access	ro
PDO Mapping	TxPDO
Value Range	Unsigned 16
Default Value	no

7.4.5.4 Object $60D8_h$: Touch probe 2 negative edge counter

This object shall provide a continuous counter that is incremented with each negative edge at touch probe 2. The counter is only valid if Touch probe input is enabled ($60B8_h$: Bit 8 = 1). Table 53 specifies the object description, and Table 54 specifies the entry description.

Table 53: Object description

Attribute	Value
Index	$60D8_h$
Name	Touch probe 2 negative edge counter
Object Code	VAR
Data Type	Unsigned16
Category	Optional

Table 54: Entry description

Attribute	Value
Sub-Index	00_h

Attribute	Value
Access	ro
PDO Mapping	TxPDO
Value Range	Unsigned 16
Default Value	no

7.4.6 Application Example: Controller based “homing” with touch probe

Figure 23 shows an example for the usage of the touch probe function.

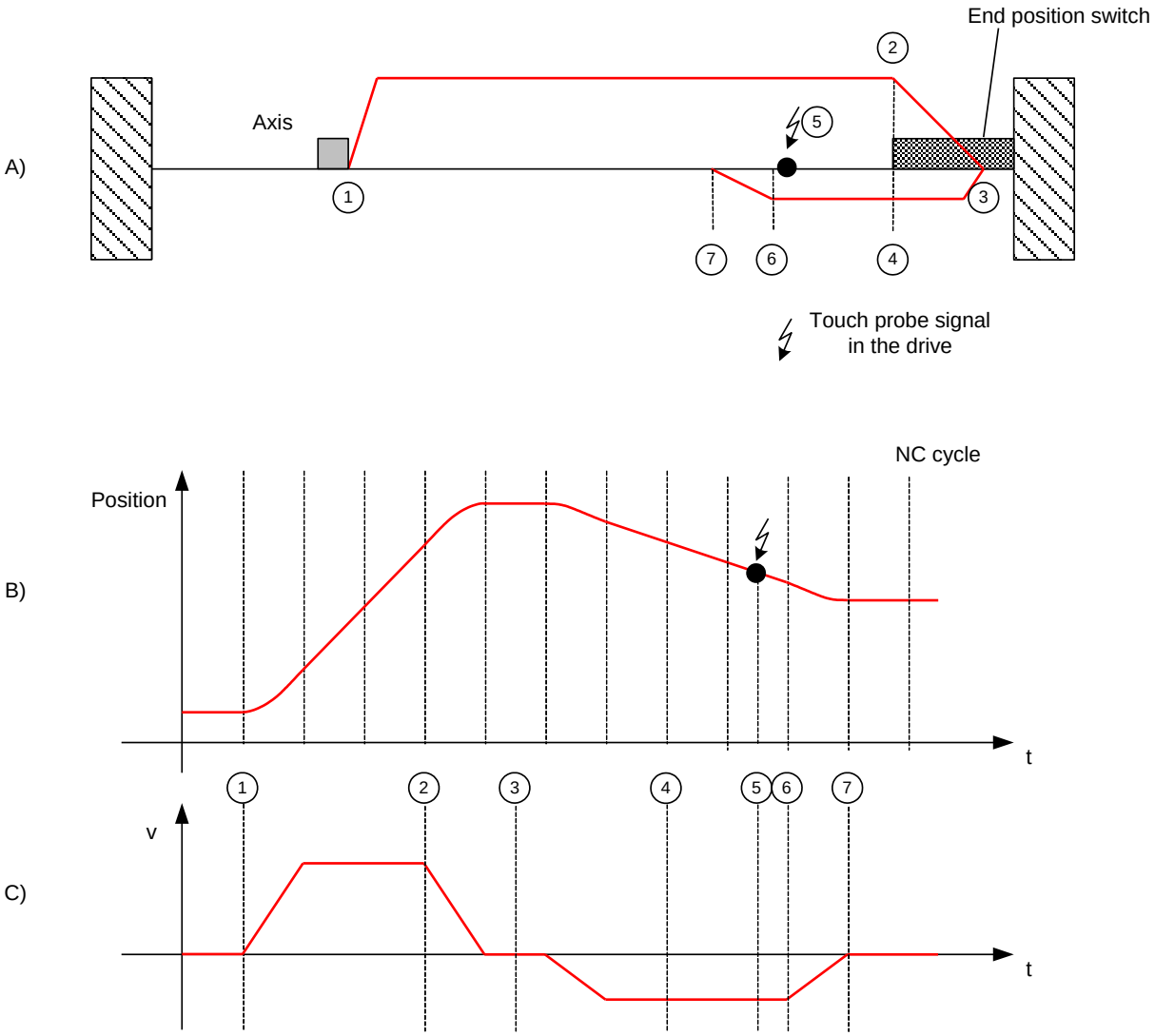


Figure 23: Control device based homing with touch probe

Part A) shows the way of the Axis within the machine. The red line denotes the velocity. In part B) the timing diagram of the position and in the part C) the timing diagram of the velocity is shown.

Table 55: Steps for control device based homing

Number	Controller action	Drive action
(1)	Accelerate the axis	
(2)	The end position switch is reached and signaled to the control device. The control device decelerates the axis to zero	

Number	Controller action	Drive action
(3)	The axis is stopped within the end position switch. The direction of the axis is changed.	
(4)	The end position switch is left. The control device enables the Touch probe function in the slave ($60B8_h$.0 = 1 for touch probe 1 or $60B8_h$.8 = 1 for touch probe 2)	Touch probe enabled
(5)	Read touch probe status ($60B9_h$) continuously	The touch probe event occurs. The slave captures the current position.
(6)	With the next cycle the control device decelerates the axis.	
(7)	The axis is stopped. The control device calculates the reference position with the captured position from the slave and the current position of the axis.	

The touch probe event is sampled locally in the drive. So it is independent of the controller (NC) cycle time and can be captured more precisely.

The detection of the end position switch doesn't need to be very accurate. It is used for changing the direction and to activate the touch probe function in the drive. A digital Input of the drive ($60FD_h$) which is mapped to the PDOs or a standard input of the control device can be used.

The touch probe position ($60BA_h$ or $60BB_h$ or $60BC_h$ or $60BD_h$) should be mapped to the PDO (see clause 11).

8 Factor Group

The Factor Group shall be used according to [3].

If the Factor Group objects are not supported by the drive the default values given in Table 56 shall be used:

Table 56: Default values for Units

Parameter	Default Unit
Position Unit	inc
Velocity Unit	inc/s
Acceleration Unit	inc/s^2
Pos Encoder resolution	2^{16} inc/rev
Torque Unit	as given in 6071_h: 0,1%

For following objects/entries the value 0 equals 2^{32}.

- $608F_h$:SI1 (Position Encoder resolution – encoder increments),
- 6090_h:SI1 (Velocity Encoder resolution – encoder increments per seconds),
- $60E6_h$:SI1-SI254 (Additional Position Encoder resolution – encoder increments)
- $60E7_h$:SI1-SI254 (Additional Velocity Encoder resolution – encoder increments per seconds)

NOTE This is defined to allow full resolution of 2^{32} (up to now only 2^{32} - 1 was possible).

9 Support of additional sensor interfaces

A drive may support several sensor interfaces. The information coming from this/these additional sensors should be given in the objects listed in Table 57.

Table 57: Additional sensor information – Object list

Index	Name	Category	Remark
$60E4_h$	Additional position actual value	O	According to 6064_h *Position actual value*
$60E5_h$	Additional velocity actual value	O	According to $606C_h$ *Velocity actual value*
$60E6_h$	Additional position encoder resolution - encoder increments	O	According to $608F_h$ *Position encoder resolution*
$60EB_h$	Additional position encoder resolution - motor revolutions	O	
$60E7_h$	Additional velocity encoder resolution - Encoder increments per second	O	According to 6090_h *Velocity encoder resolution*
$60EC_h$	Additional velocity encoder resolution - Motor revolutions per second	O	
$60E8_h$	Additional gear ratio - Motor revolutions	O	According to 6091_h *Gear ratio*
$60ED_h$	Additional gear ratio - Shaft revolutions	O	
$60E9_h$	Additional feed constant - Feed	O	According to 6092_h *Feed constant*
$60EE_h$	Additional feed constant - Shaft revolutions	O	

9.1.1 Object $60E4_h$: Additional position actual value

This object defines one or more additional position actual values. Table 58 specifies the object description, and Table 59 specifies the entry description.

Table 58: Object description object $60E4_h$

Attribute	Value
Index	$60E4_h$
Name	Additional position actual value
Object Code	ARRAY
Data Type	Integer32
Category	Optional

Table 59: Entry description object $60E4_h$

Attribute	Value
Sub-Index	00_h
Description	Highest sub-index supported
Entry category	Mandatory
Access	constant
PDO Mapping	no
Value Range	1 to 254
Default Value	Manufacturer-specific
Sub-Index	01_h
Description	1st additional position actual value
Entry category	Mandatory
Access	ro
PDO Mapping	yes
Value Range	Integer32
Default Value	No
Sub-Index	02_h
Description	2nd additional position actual value
Entry category	Optional
Access	ro
PDO Mapping	yes
Value Range	Integer32
Default Value	No
to	
Sub-Index	FE_h
Description	254th additional position actual value
Entry category	Optional
Access	ro
PDO Mapping	yes
Value Range	Integer32
Default Value	No

9.1.2 Object $60E5_h$: Additional velocity actual value

This object defines one or more additional velocity actual values. Table 60 specifies the object description, and Table 61 specifies the entry description.

Table 60: Object description object $60E5_h$

Attribute	Value
Index	$60E5_h$
Name	Additional velocity actual value
Object Code	ARRAY
Data Type	Integer32
Category	Optional

Table 61: Entry description object x60E5

Attribute	Value
Sub-Index	00_h
Description	Highest sub-index supported
Entry category	Mandatory
Access	constant
PDO Mapping	no
Value Range	1 to 254
Default Value	Manufacturer-specific
Sub-Index	01_h
Description	1st additional velocity actual value
Entry category	Mandatory
Access	ro
PDO Mapping	yes
Value Range	Integer32
Default Value	No
Sub-Index	02_h
Description	2nd additional velocity actual value
Entry category	Optional
Access	ro
PDO Mapping	yes
Value Range	Integer32
Default Value	No
to	
Sub-Index	FE_h
Description	254th additional velocity actual value
Entry category	Optional
Access	ro
PDO Mapping	yes
Value Range	Integer32
Default Value	No

9.1.3 Object $60E6_h$: Additional position encoder resolution – encoder increments

This object defines one or more additional position encoder resolution - encoder increments.

Table 62 specifies the object description, and Table 63 specifies the entry description.

Table 62: Object description object $60E6_h$

Attribute	Value
Index	$60E6_h$
Name	Additional position encoder resolution – encoder increments
Object Code	ARRAY
Data Type	Unsigned32
Category	Optional

Table 63: Entry description

Attribute	Value
Sub-Index	00_h
Description	Highest sub-index supported
Entry category	Mandatory
Access	constant
PDO Mapping	no
Value Range	1 to 254
Default Value	Manufacturer-specific
Sub-Index	01_h
Description	1st additional position encoder resolution – encoder increments
Entry category	Mandatory
Access	rw
PDO Mapping	yes
Value Range	Unsigned32 (0 equals 2^{32})
Default Value	No
Sub-Index	02_h
Description	2nd additional position encoder resolution – encoder increments
Entry category	Optional
Access	rw
PDO Mapping	yes
Value Range	Unsigned32 (0 equals 2^{32})
Default Value	No
	to
Sub-Index	FE_h
Description	254th additional position encoder resolution– encoder increments
Entry category	Optional
Access	rw
PDO Mapping	yes
Value Range	Unsigned32 (0 equals 2^{32})
Default Value	No

The calculation of the additional position encoder resolution is defined in combination with corresponding entry of object $60EB_h$ according to the definition given in object $608F_h$ *Position encoder resolution*.

9.1.4 Object $60EB_h$: Additional position encoder resolution – Motor revolutions

This object defines one or more additional position encoder resolution – Motor revolutions.

Table 64 specifies the object description, and Table 65 specifies the entry description.

Table 64: Object description object $60EB_h$

Attribute	Value
Index	$60E6_h$
Name	Additional position encoder resolution – Motor revolutions
Object Code	ARRAY
Data Type	Unsigned32
Category	Optional

Table 65: Entry description

Attribute	Value
Sub-Index	00_h
Description	Highest sub-index supported
Entry category	Mandatory
Access	constant
PDO Mapping	no
Value Range	1 to 254
Default Value	Manufacturer-specific
Sub-Index	01_h
Description	1st additional position encoder resolution – Motor revolutions
Entry category	Mandatory
Access	rw
PDO Mapping	yes
Default Value	No
Sub-Index	02_h
Description	2nd additional position encoder resolution – Motor revolutions
Entry category	Optional
Access	rw
PDO Mapping	yes
Default Value	No
	to
Sub-Index	FE_h
Description	254th additional position encoder resolution– Motor revolutions
Entry category	Optional
Access	rw
PDO Mapping	yes
Default Value	No

The calculation of the additional position encoder resolution is defined in combination with corresponding entry of object $60E6_h$ according to the definition given in object $608F_h$ *Position encoder resolution*.

9.1.5 Object $60E7_h$: Additional velocity encoder resolution - Encoder increments per second

This object defines one or more additional velocity encoder resolution - Encoder increments per second. Table 66 specifies the object description, and Table 67 specifies the entry description.

Table 66: Object description object $60E7_h$

Attribute	Value
Index	$60E7_h$
Name	Additional velocity encoder resolution - Encoder increments per second
Object Code	ARRAY
Data Type	Unsigned32
Category	Optional

Table 67: Entry description

Attribute	Value
Sub-Index	00_h
Description	Highest sub-index supported
Entry category	Mandatory
Access	constant
PDO Mapping	no
Value Range	1 to 254
Default Value	Manufacturer-specific
Sub-Index	01_h
Description	1st additional velocity encoder resolution - Encoder increments per second
Entry category	Mandatory
Access	rw
PDO Mapping	yes
Value Range	Unsigned32 (0 equals 2^{32})
Default Value	No
Sub-Index	02_h
Description	2nd additional velocity encoder resolution - Encoder increments per second
Entry category	Optional
Access	rw
PDO Mapping	yes
Value Range	Unsigned32 (0 equals 2^{32})
Default Value	No
	to
Sub-Index	FE_h
Description	254th additional velocity encoder resolution - Encoder increments per second
Entry category	Optional
Access	rw
PDO Mapping	yes
Value Range	Unsigned32 (0 equals 2^{32})

Attribute	Value
Default Value	No

The calculation of the additional velocity encoder resolution is defined in combination with corresponding entry of object $60EC_h$ according to the definition given in object 6090_h *Velocity encoder resolution*.

9.1.6 Object $60EC_h$: Additional velocity encoder resolution – Motor revolutions per second

This object defines one or more additional velocity encoder resolution - Motor revolutions per second. Table 68 specifies the object description, and Table 69 specifies the entry description.

Table 68: Object description object $60EC_h$

Attribute	Value
Index	$60E7_h$
Name	Additional velocity encoder resolution - Motor revolutions per second
Object Code	ARRAY
Data Type	Unsigned32
Category	Optional

Table 69: Entry description

Attribute	Value
Sub-Index	00_h
Description	Highest sub-index supported
Entry category	Mandatory
Access	constant
PDO Mapping	no
Value Range	1 to 254
Default Value	Manufacturer-specific
Sub-Index	01_h
Description	1st additional velocity encoder resolution - Motor revolutions per second
Entry category	Mandatory
Access	rw
PDO Mapping	yes
Default Value	No
Sub-Index	02_h
Description	2nd additional velocity encoder resolution - Motor revolutions per second
Entry category	Optional
Access	rw
PDO Mapping	yes
Default Value	No
	to
Sub-Index	FE_h
Description	254th additional velocity encoder resolution - Motor revolutions per second
Entry category	Optional
Access	rw

Attribute	Value
PDO Mapping	yes
Default Value	No

The calculation of the additional velocity encoder resolution is defined in combination with corresponding entry of object $60E7_h$ according to the definition given in object 6090_h *Velocity encoder resolution*.

9.1.7 Object $60E8_h$: Additional gear ratio – Motor revolutions

This object defines one or more additional gear ration – Motor revolutions. Table 70 specifies the object description, and Table 71 specifies the entry description.

Table 70: Object description object $60E8_h$

Attribute	Value
Index	$60E8_h$
Name	Additional gear ratio – Motor revolutions
Object Code	ARRAY
Data Type	Unsigned32
Category	Optional

Table 71: Entry description

Attribute	Value
Sub-Index	00_h
Description	Highest sub-index supported
Entry category	Mandatory
Access	constant
PDO Mapping	no
Value Range	1 to 254
Default Value	Manufacturer-specific
Sub-Index	01_h
Description	1st additional gear ratio – Motor revolutions
Entry category	Mandatory
Access	rw
PDO Mapping	yes
Default Value	No
Sub-Index	02_h
Description	2nd additional gear ratio – Motor revolutions
Entry category	Optional
Access	rw
PDO Mapping	yes
Default Value	No
to	
Sub-Index	FE_h
Description	254th additional gear ratio – Motor revolutions
Entry category	Optional
Access	rw

Attribute	Value
PDO Mapping	yes
Default Value	No

The calculation of the additional gear ratio is defined in combination with corresponding entry of object $60ED_h$ according to the definition given in object 6091_h *Gear ratio*.

9.1.8 Object $60ED_h$: Additional gear ratio – Shaft revolutions

This object defines one or more additional gear ration – Shaft revolutions. Table 72 specifies the object description, and Table 73 specifies the entry description.

Table 72: Object description object $60ED_h$

Attribute	Value
Index	$60E8_h$
Name	Additional gear ratio – Shaft revolutions
Object Code	ARRAY
Data Type	Unsigned32
Category	Optional

Table 73: Entry description

Attribute	Value
Sub-Index	00_h
Description	Highest sub-index supported
Entry category	Mandatory
Access	constant
PDO Mapping	no
Value Range	1 to 254
Default Value	Manufacturer-specific
Sub-Index	01_h
Description	1st additional gear ratio – Shaft revolutions
Entry category	Mandatory
Access	rw
PDO Mapping	yes
Default Value	No
Sub-Index	02_h
Description	2nd additional gear ratio – Shaft revolutions
Entry category	Optional
Access	rw
PDO Mapping	yes
Default Value	No
to	
Sub-Index	FE_h
Description	254th additional gear ratio – Shaft revolutions
Entry category	Optional
Access	rw
PDO Mapping	yes
Default Value	No

The calculation of the additional gear ratio is defined in combination with corresponding entry of object $60E8_h$ according to the definition given in object 6091_h *Gear ratio*.

9.1.9 Object $60E9_h$: Additional feed constant – Feed

This object defines one or more additional feed constant values – Feed. Table 74 specifies the object description, and Table 75 specifies the entry description.

Table 74: Object description object $60E9_h$

Attribute	Value
Index	$60E9_h$
Name	Additional feed constant – Feed
Object Code	ARRAY
Data Type	Unsigned32
Category	Optional

Table 75: Entry description

Attribute	Value
Sub-Index	00_h
Description	Highest sub-index supported
Entry category	Mandatory
Access	Constant
PDO Mapping	No
Value Range	1 to 254
Default Value	Manufacturer-specific
Sub-Index	01_h
Description	1st additional feed constant – Feed
Entry category	Mandatory
Access	Rw
PDO Mapping	Yes
Default Value	No
Sub-Index	02_h
Description	2nd additional feed constant – Feed
Entry category	Optional
Access	Rw
PDO Mapping	Yes
Default Value	No
To	
Sub-Index	FE_h
Description	254th additional feed constant – Feed
Entry category	Optional
Access	Rw
PDO Mapping	Yes
Default Value	No

The calculation of the additional feed constant is defined in combination with corresponding entry of object $60EE_h$ according to the definition given in object 6092_h *Feed constant*.

9.1.10 Object $60EE_h$: Additional feed constant – Shaft revolutions

This object defines one or more additional feed constant values – Shaft revolutions. Table 76 specifies the object description, and Table 77 specifies the entry description.

Table 76: Object description object $60EE_h$

Attribute	Value
Index	$60E9_h$
Name	Additional feed constant – Shaft revolutions
Object Code	ARRAY
Data Type	Unsigned32
Category	Optional

Table 77: Entry description

Attribute	Value
Sub-Index	00_h
Description	Highest sub-index supported
Entry category	Mandatory
Access	constant
PDO Mapping	no
Value Range	1 to 254
Default Value	Manufacturer-specific
Sub-Index	01_h
Description	1st additional feed constant – Shaft revolutions
Entry category	Mandatory
Access	rw
PDO Mapping	yes
Default Value	No
Sub-Index	02_h
Description	2nd additional feed constant – Shaft revolutions
Entry category	Optional
Access	rw
PDO Mapping	yes
Default Value	No
to	
Sub-Index	FE_h
Description	254tn additional feed constant – Shaft revolutions
Entry category	Optional
Access	rw
PDO Mapping	yes
Default Value	No

The calculation of the additional feed constant is defined in combination with corresponding entry of object $60E9_h$ according to the definition given in object 6092_h *Feed constant*.

10 Endless positioning

The position data range (607A$_h$ Target position, 6064$_h$ Position actual value) and the machine relevant absolute position data of the axis' sensor may differ. If the position data of an axis with infinite travel range (e.g. rotary axis, spindle etc.) were processed in absolute format, the axis would risk moving beyond the value range of the position data. This would lead to invalid position data; operating modes with position control would not be safe to operate.

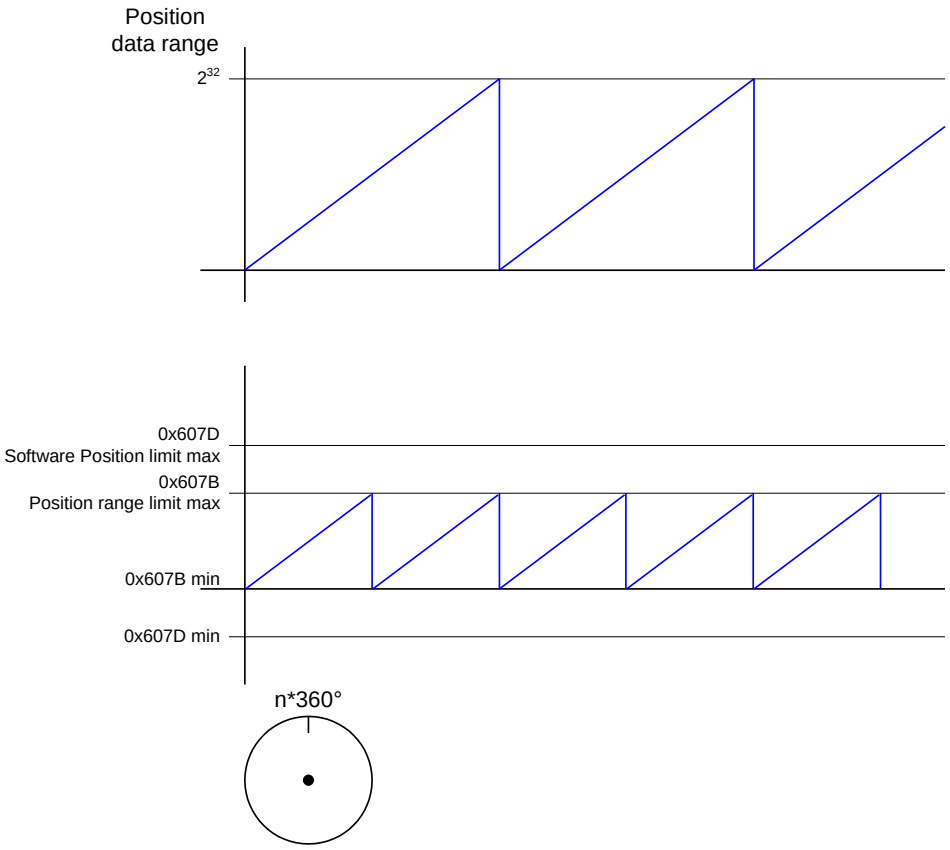


Figure 24: Position actual value in modulo format

The object 607B$_h$ Position range limit indicates the maximum and minimum position range limits. It shall limit the range of the input value. On reaching or exceeding these limits, the input value shall wrap automatically to the other end of range. The Software position limit 607D$_h$ shall be out of the range of the position range limits.

To disable the position range limits for any reason the min position range limit (607B$_h$:01) and max position range limit (607B$_h$:02) shall be set to 0.

The drive shall always use the shortest way to the next Target position. Therefore the position command difference between two consecutive cycles may not exceed half the range:

Maximum position command difference = Position range / 2

Different movements are defined in object Positioning option code 60F2$_h$ in the bits 6 and 7.

Possible movements:

- normal positioning
- positioning only in negative direction
- positioning only in positive direction
- positioning with the shortest way
- movement greater than a modulo value

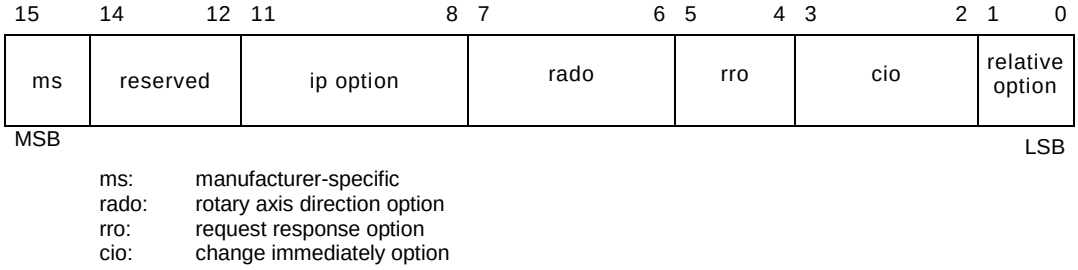


Figure 25: Object structure 60F2$_h$

Table 78 shows the bit value definition of bit 6 and 7.

Table 78: Value definition for bit 6 and 7 of object 60F2$_h$

Bit 7	Bit 6	Definition
0	0	Normal positioning similar to linear axis If reaching or exceeding the position range limits (607B$_h$) the input value shall wrap automatically to the other end of the range
0	1	Positioning only in negative direction; if target position is higher than actual position, axis moves over “Min position limit“ to target position
1	0	Positioning only in positive direction; if target position is lower than actual position, axis moves over “Max position limit“ to target position
1	1	Positioning with the shortest way to the target position. Special condition: If the difference between actual value and target position in a 360° system is 180°, the axis will move in positive direction.

Figure 26 depicts the movements depending on the settings of the rotary axis direction option bits. In this example min position range limit is 0 and max position range limit is 360.

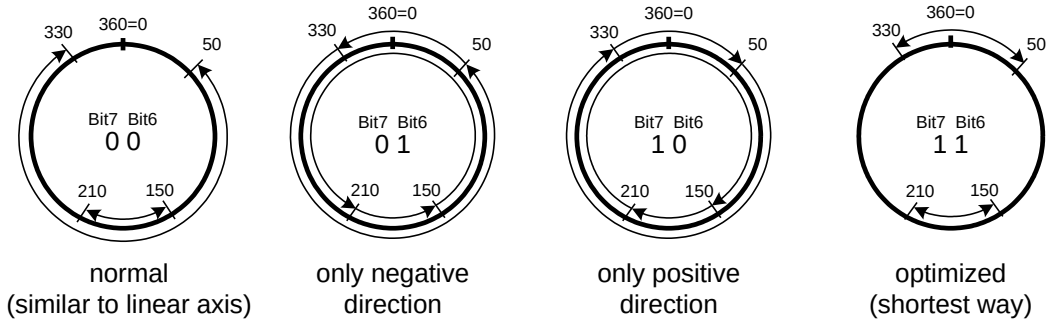


Figure 26: Rotary axis positioning

11 Process data objects (PDO)

The mapping of the process data is not fixed by this directive, but defined by the device description of the drive.

The following PDO mappings are recommended.

Table 79: Default PDOs

Direction	Description
Operation mode csp	
RPDO	Controlword (6040_h), Target position ($607A_h$)
TPDO	Statusword (6041_h), Position actual value (6064_h)
Operation mode csv	
RPDO	Controlword (6040_h), Target velocity ($60FF_h$)
TPDO	Statusword (6041_h), Position actual value (6064_h)
Operation mode cst	
RPDO	Controlword (6040_h), Target torque (6071_h)
TPDO	Statusword (6041_h), Position actual value (6064_h), Torque actual value (6077_h)
Operation mode csp, csv, cst	
RPDO	Controlword (6040_h), Target position ($607A_h$), Target velocity ($60FF_h$), Target torque (6071_h), Max Torque (6072_h), Modes of operation (6060_h), Touch probe function ($60B8_h$)
TPDO	Statusword (6041_h), Position actual value (6064_h), Torque actual value (6077_h), Following error actual value ($60F4_h$), Modes of operation display (6061_h), Touch probe status ($60B9_h$), Touch probe value*

* NOTE: The object of the touch probe value depends on the configured touch probe input. It can be $60BA_h$, $60BB_h$, $60BC_h$ or $60BD_h$.

It is optional to add more objects to the proposed PDOs and a mixture of the PDOs can be configured without repeating same objects like Controlword or Statusword.

The default mapping for a drive is defined in the device description file.

11.1 Example of PDO mapping implementations for dynamic switching of operation modes

For a dynamic switching of operation mode in state operation the drive shall support the “Modes of Operation” and “Modes of operation Display” objects, see 6.9.2. These objects should be part of the PDO Mapping.

Table 80 shows which objects should be mapped according to the supported modes of operations.

Table 80: Recommended PDO Mapping according to supported modes of operation

Index	Name	csp only	csv only	cst only	csp + csv	csp + cst	csv + cst	csp + csv + cst	pp + hm
RxPDOs									
6040_h	Controlword	+	+	+	+	+	+	+	+
6060_h	Modes of operation	-	-	-	+	+	+	+	o
$607A_h$	Target Position	+	-	-	+	+	-	+	+
$60FF_h$	Target velocity	-	+	-	+	-	+	+	-
6071_h	Target Torque	-	-	+	-	+	+	+	-
TxPDOs									
6041_h	Statusword	+	+	+	+	+	+	+	+
6061_h	Modes of operation display	-	-	-	+	+	+	+	o
6064_h	Position actual value	+	+	+	+	+	+	+	+
6077_h	Torque actual value	-	-	+	-	+	+	+	-

+ = should = recommended
0 = can = possible
- = not used

12 Synchronization

There are no further specifications in this directive regarding synchronization.

Synchronization and error handling are described in [9].

For the Drive

- DC synchronous mode is very useful for csp, csv and cst mode. It is recommended to support this synchronization mode.
- Specific values for synchronization (e.g. shift time, internal delay times) are specified in [11].

For the control device

- control device shall provide new set points each bus cycle.

13 Stepper Motor specifics

13.1 Basic assumption for stepper drives in this directive

Stepper drives can be used with or without a feedback system.

- Stepper drives with feedback system can be used like servo drive since the control unit can make use of the position feedback.
- Usage of stepper drives without feedback system is described within this chapter

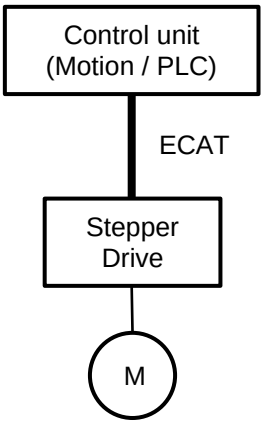


Figure 27: Stepper drive without feedback system

13.2 Resolution factor

Typically, servo drives with position feedback are configured with the object 6091_h *Gear ratio*, 6092_h *Feed constant* and $608F_h$ *Position encoder resolution*.

Stepper motor does not provide *Position encoder resolution*, but uses *Motor resolution* instead.

Stepper motors come with a hardware defined resolution, i.e. full steps per revolution. Stepper drives can calculate additional micro steps for each full step of the motor. The Motor resolution defines the combination of full steps and micro steps.

EXAMPLE: The stepper motor comes with 200 full steps (1,8°). The drive can control the motor with 2048 micro steps per full step. This leads to a resolution of 409600 micro steps / revolution

The Motor resolution can be calculated with following formula:

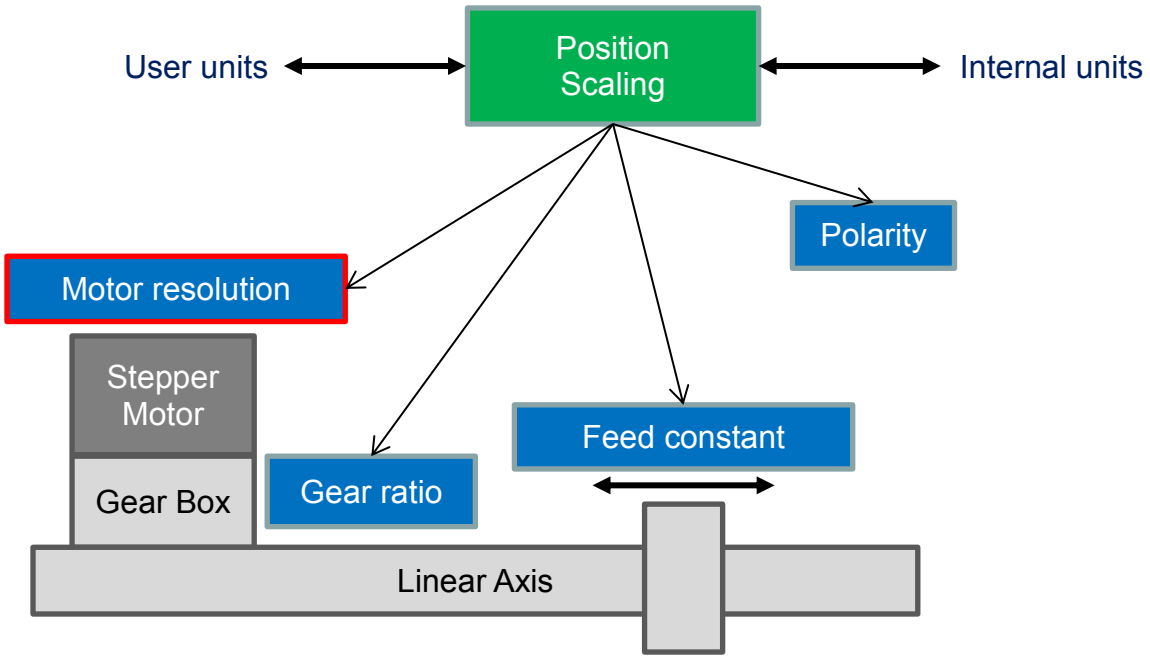
$$Motor\ resolution = \frac{fullsteps * micro\ steps}{revolution}\ [steps\ /\ revolution]$$

The step angle is given in

$$step\ angle = \frac{360°}{fullsteps}$$

Corresponding to the calculation of position scaling for servo drives in [3] the calculation of the position scaling is done by the following formula:

$$position\ acutal\ value = \frac{position\ internal\ value\ \times\ feed\ constant}{Motor\ resolution\ \times Gear\ Ratio}$$



13.2.1 Object 60EF$_h$: Motor resolution

This object defines the Motor resolution of a stepper motor in steps / revolution.

Table 81 specifies the object description, and Table 82 specifies the entry description.

Table 81: Object description object 60EF$_h$

Attribute	Value
Index	60EF$_h$
Name	Motor resolution [steps / revolution]
Object Code	VAR
Data Type	Unsigned32
Category	Optional

Table 82: Entry description

Attribute	Value
Sub-Index	00$_h$
Access	ro
PDO Mapping	no
Value Range	Unsigned32
Default Value	Manufacturer-specific

14 Guidance for drive use cases

According to the control unit capabilities and the supported modes of operation in a drive different use cases are commonly used.

PLCs often are designed to not support a cyclic synchronous behavior but to support modes such as point- to-point movement of the drive.

Motion controllers are expected to support cyclic synchronous closed loop control.

Table 83: Drive use cases regarding to PLC functionality

Use Case	Control Unit	Drive: supported modes of operation	Remarks
A	PLC or Motion controller	csp + pp/hm + csv + cst	Full functionality for servo drives
B	PLC or Motion controller	csp + pp/hm	Full functionality for stepper drives with position feedback
C	Motion controller	csp	**Basic use case for CiA402 drives.** Simple functionality for servo drives
D	PLC	pp/hm	Simple functionality for stepper drives without position feedback

15 Device Type

The object 1000_h *Device Type* indicates the type of device and its functionality. For the EtherCAT CiA402 servo drive the Device Type in object 1000_h is set as shown in Table 84.

Bit 19 in the Type sub-field of the Additional Information field shall be set to 1_b, when the device complies with the Safety Drive Profile defined in ETG.6100 series [10].

Table 84: Object 1000_h Device Type for EtherCAT servo drive

Object 1000_h	**Additional Information**		**Device Profile Number**	
	31 24	23 16	15 0	
	Mode Bits	Type	Device Profile Number	
EtherCAT servo drive no safety option	*	$0000\ 0010_b$ $=02_h$	0192_h (402_d)	
EtherCAT servo drive supporting ETG.6100 Safety Drive Profile	*	$0000\ 1010_b$ $=0A_h$	0192_h (402_d)	
EtherCAT Stepper drive no safety option	*	$0000\ 0100_b$ $=04_h$	0192_h (402_d)	
EtherCAT Stepper drive supporting ETG.6100 Safety Drive Profile	*	$0000\ 1100_b$ $=0C_h$	192_h (402_d)	

* means manufacturer specific

The Type = 02_h is defined in [7] for generic servo drive devices. Even though the standard defines a PDO set for generic servo drives this not adopted for an EtherCAT servo.

16 Object list

For multi axis devices the object range 6000_h to $67FF_h$ shall be shifted by 800_h per axis as described in the IEC 61800-7-201.

Table 85: Object dictionary – object list (numerical order)

Index	Name	Category	Remark
1000_h	Device Type	M	supporting generic PDO mapping = $xx020192_h$
1001_h	Error Register	M	
1018_h	Identity object	M	
6007_h	Abort connection option code	O	
$603D_h$	Reserved (RxPDO State)	O	
$603E_h$	TxPDO State	O	
$603F_h$	Error Code	O	
6040_h	Controlword	M	
6041_h	Statusword	M	
6042_h	vl target velocity	C	Mandatory if vl is supported
6043_h	vl velocity demand	C	Mandatory if vl is supported
6044_h	vl velocity actual value	C	Mandatory if vl is supported
6045_h	reserved	-	reserved for compatibility reasons
6046_h	vl velocity min max amount	C	Mandatory if vl is supported
6047_h	reserved	-	reserved for compatibility reasons
6048_h	vl velocity acceleration	C	Mandatory if vl is supported
6049_h	vl velocity deceleration	O	
$604A_h$	vl velocity quick stop	O	
$604B_h$	vl set-point factor	O	
$604C_h$	vl dimension factor	O	
$604D_h$	reserved	-	reserved for compatibility reasons
$604E_h$	reserved	-	reserved for compatibility reasons
$604F_h$	reserved	-	reserved for compatibility reasons
6052_h … 6059_h	reserved	-	reserved for compatibility reasons
$605A_h$	Quick stop option code	O	
$605B_h$	Shutdown option code	O	
$605C_h$	Disable operation option code	O	
$605D_h$	Halt option code	O	
$605E_h$	Fault reaction option code	O	
6060_h	Modes of operation	M	Mandatory
6061_h	Modes of operation display	M	Mandatory
6062_h	Position demand value	O	
6063_h	Position actual internal value	O	
6064_h	Position actual value	C	Mandatory, if csp or csv is supported Recommended, if cst is supported
6065_h	Following error window	C	Recommended, if csp is supported
6066_h	Following error time out	C	Recommended, if csp is supported
6067_h	Position window	O	
6068_h	Position window time	O	
6069_h	Velocity sensor actual value	O	
$606A_h$	Sensor selection code	O	
$606B_h$	Velocity demand value	O	

Index	Name	Category	Remark
$606C_h$	Velocity actual value	C	Mandatory if pv supported Recommended if csv is supported
$606D_h$	Velocity window	O	
$606E_h$	Velocity window time	O	
$606F_h$	Velocity threshold	O	
6070_h	Velocity threshold time	O	
6071_h	Target Torque	C	Mandatory if tq or cst is supported
6072_h	Max torque	C	Mandatory if FG Torque Limiting is supported
6073_h	Max current	O	
6074_h	Torque demand	O	
6075_h	Motor rated current	O	
6076_h	Motor rated torque	O	
6077_h	Torque actual value	C	Mandatory if cst is supported Recommended if csp or csv is supported
6078_h	Current actual value	O	
6079_h	DC link circuit voltage	O	
$607A_h$	Target position	C	Mandatory if csp is supported
$607B_h$	Position range limit	C	Recommended if csp is supported
$607C_h$	Home offset	R	Recommended if Homing is supported
$607D_h$	Software position limit	C	Recommended if csp is supported
$607E_h$	Polarity	O	
$607F_h$	Max profile velocity	O	
6080_h	Max motor speed	O	
6081_h	Profile velocity	C	Mandatory if pp is supported
6082_h	End velocity	O	
6083_h	Profile acceleration	C	Mandatory if pp or pv is supported
6084_h	Profile deceleration	O	
6085_h	Quick stop deceleration	O	
6086_h	Motion profile type	O	
6087_h	Torque slope	C	Mandatory if tq is supported
6088_h	Torque profile type	O	
6089_h …$608E_h$	reserved	-	reserved for compatibility reasons
$608F_h$	Position encoder resolution	O	
6090_h	Velocity encoder resolution	O	
6091_h	Gear ratio	O	
6092_h	Feed constant	O	
6093_h …6097_h	reserved	-	reserved for compatibility reasons
6098_h	Homing method	C	Mandatory if Homing is supported
6099_h	Homing speed	O	
$609A_h$	Homing acceleration	O	
$60A0_h$	reserved	-	reserved for compatibility reasons
$60A1_h$	reserved	-	reserved for compatibility reasons
$60A3_h$	Profile jerk use	O	
$60A4_h$	Profile jerk	O	
$60B0_h$	Position offset	O	
$60B1_h$	Velocity offset	O	Recommended if csp is supported

Index	Name	Category	Remark
$60B2_h$	Torque offset	O	Recommended if csp or csv is supported
$60B8_h$	Touch probe function	C	Mandatory if FG Touch Probe is supported
$60B9_h$	Touch probe status	C	Mandatory if FG Touch Probe is supported
$60BA_h$	Touch probe position 1 positive value	C	Mandatory if FG Touch Probe is supported
$60BB_h$	Touch probe position 1 negative value	C	Mandatory if FG Touch Probe is supported
$60BC_h$	Touch probe position 2 positive value	O	Optional if FG Touch Probe and Touch probe input 2 is supported
$60BD_h$	Touch probe position 2 negative value	O	Optional if FG Touch Probe and Touch probe input 2 is supported
$60C0_h$	Interpolation sub mode select	O	
$60C1_h$	Interpolation data record	O	
$60C2_h$	Interpolation time period	C	Mandatory if ip mode is supported Recommended if csp, csv or cst is supported
$60C4_h$	Interpolation data configuration	O	
$60C5_h$	Max acceleration	O	
$60C6_h$	Max deceleration	O	
$60D0_h$	Touch probe source	C	Recommended if FG Touch probe is supported
$60D1_h$	Touch probe time stamp 1 positive value	O	
$60D2_h$	Touch probe time stamp 1 negative value	O	
$60D3_h$	Touch probe time stamp 2 positive value	O	
$60D4_h$	Touch probe time stamp 2 negative value	O	
$60D5_h$	Touch probe 1 positive edge counter	O	
$60D6_h$	Touch probe 1 negative edge counter	O	
$60D7_h$	Touch probe 2 positive edge counter	O	
$60D8_h$	Touch probe 2 negative edge counter	O	
$60D9_h$	Supported synchronization functions	O	
$60DA_h$	Synchronization Function settings	O	
$60E0_h$	Positive torque limit value	C	Mandatory if FG Torque Limiting is supported
$60E1_h$	Negative torque limit value	C	Mandatory if FG Torque Limiting is supported
$60E3_h$	Supported homing methods	C	Recommended if Homing is supported
$60E4_h$	Additional position actual value	O	
$60E5_h$	Additional velocity actual value	O	
$60E6_h$	Additional position encoder resolution – encoder increments	O	

Index	Name	Category	Remark
$60E7_h$	Additional velocity encoder resolution - Encoder increments per second	O	
$60E8_h$	Additional gear ratio – Motor revolutions	O	
$60E9_h$	Additional feed constant - Feed	O	
$60EA_h$	Commutation angle	C	Mandatory if *cstca* mode supported
$60EB_h$	Additional position encoder resolution – motor revolutions	O	
$60EC_h$	Additional velocity encoder resolution – Motor revolutions per second	O	
$60ED_h$	Additional gear ratio – Shaft revolutions	O	
$60EE_h$	Additional feed constant – Shaft revolutions	O	
$60EF_h$	Motor resolution	O	
$60F2_h$	Position option code	O	
$60F4_h$	Following error actual value	C	Recommended if csp is supported
$60F6_h$	reserved	-	reserved for compatibility reasons
$60F7_h$	reserved	-	reserved for compatibility reasons
$60F8_h$	Max slippage	O	
$60F9_h$	reserved	-	reserved for compatibility reasons
$60FA_h$	Control effort	O	
$60FB_h$	reserved	-	reserved for compatibility reasons
$60FC_h$	Position demand internal value	O	
$60FD_h$	Digital inputs	O	
$60FE_h$	Digital outputs	O	
$60FF_h$	Target velocity	C	Mandatory if pv or csv is supported
6200_h … $62FF_h$	Profile 452 PLCopen Motion Control	O	
6402_h	Motor Type	O	
6403_h	Motor catalogue number	O	
6404_h	Motor manufacturer	O	
6405_h	http motor catalogue address	O	
6406_h	Motor calibration date	O	
6407_h	Motor service period	O	
6410_h	reserved	-	reserved for compatibility reasons
6502_h	Supported drive modes	M	
6503_h	Drive catalogue number	O	
6504_h	reserved	-	reserved for compatibility reasons
6505_h	http drive catalogue address	O	
6510_h	reserved	-	reserved for compatibility reasons
6600_h … $67EF_h$	Safety Drive Profile	O	See ETG.6100
$67F0_h$ … $67FE_h$	reserved		
$67FF_h$	Device profile number (for multiple device module)	O	